

SINGULAR POINTS OF KLT DEL PEZZO SURFACES OF PICARD NUMBER ONE IN POSITIVE CHARACTERISTIC

ABSTRACT. Let X be a normal projective surface over an algebraically closed field of odd positive characteristic. Suppose that X has Kawamata log terminal singularities, that $-K_X$ is ample and $\mathbb{Q}$ -Cartier, and that its Picard number is one. We prove that X has at most seven distinct singular points. A construction by Bernasconi gives such a surface with seven singular points in characteristic three, so the bound is sharp in that characteristic. The proof proceeds on the minimal resolution of a hypothetical counterexample. Among its smooth rational curves of self-intersection -1 , we choose one of least degree with respect to the pullback of $-K_X$. Effective divisors obtained from Riemann–Roch, together with rational fibrations, reduce the possible configurations of curves contracted by the resolution to a finite list. Intersection theory and branched double covers exclude each configuration. We also describe a construction using the Frobenius morphism that recovers Bernasconi’s surface and the characteristic-two families of Keel and McKernan. These characteristic-two constructions give surfaces with Kawamata log terminal singularities, ample anticanonical divisor, Picard number one, and arbitrarily many singular points.

1. INTRODUCTION

A *klt del Pezzo surface* over an algebraically closed field k is a normal projective surface X with Kawamata log terminal singularities such that $-K_X$ is an ample $\mathbb{Q}$ -Cartier divisor. Here Kawamata log terminal, abbreviated klt, means that all discrepancies of the pair $(X, 0)$ are greater than -1 . The Picard number $\rho(X)$ is the dimension of the real vector space of Cartier divisor classes modulo numerical equivalence. We use *rank one* only to mean Picard number one. The assumption $\rho(X) = 1$ restricts the numerical geometry of X , but does not directly bound the number of irreducible curves appearing on a resolution of its singularities. We consider the number

$$n(X) := \# \operatorname{Sing}(X)$$

of distinct singular points, rather than the number of exceptional curves or the length of a singular subscheme.

Theorem 1.1 (Uniform seven-point bound). *Let k be an algebraically closed field of characteristic $p > 2$, and let X be a normal projective surface over k . Suppose that X has klt singularities, $-K_X$ is ample and $\mathbb{Q}$ -Cartier, and $\rho(X) = 1$. Then $n(X) \leq 7$.*

There is no lower bound on the discrepancies in this statement other than the strict inequalities in the definition of klt. In particular, no fixed Cartier index is assumed. The characteristic-three case of Theorem 1.1 is sharp. In characteristic two the number of singular points is unbounded. The existence statements below are due to Bernasconi [2, Section 3.1 and Proposition 3.2] and Keel–McKernan [7, Section 9], respectively.

Theorem 1.2 (Bernasconi; Keel–McKernan). *Over every algebraically closed field of characteristic three, there is a klt del Pezzo surface X with $\rho(X) = 1$, $n(X) = 7$, and $K_X^2 = 1/3$. Over every algebraically closed field of characteristic two, for each integer $n \geq 3$ there is a klt del Pezzo surface $X_{2,n}$ with*

$$\rho(X_{2,n}) = 1, \quad n(X_{2,n}) = 2n + 1, \quad K_{X_{2,n}}^2 = \frac{2}{n-2}.$$

1.1. Related work. Over $\mathbb{C}$, Keel and McKernan obtained singular-point estimates from their study of rational curves and orbifold inequalities [7, Section 9]. Belousov proved that a

klt del Pezzo surface of Picard number one has at most four singular points [1, Theorem 1.1]. Liu and Xie subsequently established the inequality

$$n(X) \leq 2\rho(X) + 2$$

for complex klt Fano surfaces of arbitrary Picard number [10, Theorem 1.2]. These results concern characteristic zero.

In positive characteristic, Lacini's classification of rank-one log del Pezzo surfaces in characteristics different from two and three gives $n(X) \leq 4$ for $p > 5$ [9, Corollary 1.2]. Lacini also constructs a surface with five singular points in characteristic five [9, Theorem 1.3 and Example 7.6]. Nagaoka's classification of the characteristic-five surfaces whose resolutions with exceptional boundary do not lift to the ring of Witt vectors gives a more detailed description of the exceptional behavior in that characteristic [11]. His later review re-examines the classification arguments and their characteristic restrictions, and identifies a possible omission from one of Lacini's lists [12]. The proof below does not use these classifications. For $p > 5$, the known four-point bound is stronger than seven.

Keel and McKernan constructed rank-one klt del Pezzo surfaces in characteristic two with $2n+1$ singular points for arbitrarily large n [7, Section 9]. Bernasconi constructed a seven-point surface in characteristic three in his study of the failure of Kawamata–Viehweg vanishing [2, Section 3.1]; the same surface is recorded in [9, Example 7.4]. Its minimal exceptional divisor consists of four isolated (-3) -curves and three chains of two (-2) -curves. The graph-of-Frobenius construction used in section 10 is Bernasconi's construction when $p = 3$ and $n = 3$, up to interchanging the factors of $\mathbb{P}^1 \times \mathbb{P}^1$. Bernasconi also noted the reversal of the canonical sign for its higher-characteristic analogues [2, Remark 3.3]. Section 10 records a uniform canonical-divisor formula and a contraction proof using an explicit nef divisor with its exact null locus. The further calculations describe the shortest curves and their birational replacements.

Palka and Pelka pursue a classification over arbitrary algebraically closed fields by means of the *height*: the least intersection number of the reduced exceptional divisor with a fiber of a rational ruling on the minimal resolution, with value $+\infty$ if there is no such ruling. Their first paper treats heights one and two and surfaces admitting specified birational morphisms to canonical del Pezzo surfaces [13]; the next treats height three [14]. In [13, Remark 1.7], Palka and Pelka already announce that the maximal numbers of singular points in characteristics three and five are seven and five, respectively, with proofs deferred to subsequent articles. The proof below establishes the seven-point bound in every odd positive characteristic.

The operation of removing a contacted exceptional (-2) -curve and contracting a nonexceptional (-1) -curve, under a two-contact hypothesis, is an elementary swap in [13, Definition 1.3 and Lemma 2.9]. The replacements in section 4 include these operations. We require a formulation with explicit discrepancy coefficients and exact changes in the number of singular points, also allowing the additional contact configurations arising at a curve of least degree.

Dolgachev, Mendes Lopes, and Pardini associate a binary code to disjoint (-2) -curves by recording which sums are twice a Picard class, and use coverings to constrain nodal rational surfaces [5, Sections 2–3]. In particular, the fact that every such sum contains a multiple of four curves is the standard Riemann–Roch parity restriction. Calabri, Ciliberto, and Mendes Lopes show that an even set of four nodes on a smooth rational surface in characteristic different from two forces a rational fibration with two distinguished fibers [3, Theorem 1.1]. Here the double-cover argument keeps track of the entire exceptional forest, including the components outside the branch divisor, and combines its rank with the Picard-number-one hypothesis. The characteristic-two divisibility calculation identifies the classical code $DE(n)$ of [5, Section 2] on the explicit Frobenius surface.

We work on one fixed minimal resolution and choose a curve of least degree with respect to the pullback of the anticanonical divisor. Effective integral adjoints give degree inequalities without a vanishing theorem, and reduce the problem to a small classification of weighted forests rather than a classification of surfaces. The external contraction results are Keel's

semiampness criterion [6, Theorem 0.2] and the smooth-surface case of Tanaka’s contraction theorem [16, Theorem 4.4]. Within positive characteristic, the remaining restriction is precisely the separability of two degree-two constructions. Consequently the characteristic-three proof works unchanged for each fixed $p > 2$; there is no specialization or lifting argument from characteristic three.

1.2. Outline of the proof. Let $\pi : S \rightarrow X$ be the minimal resolution, let $D = \text{Exc}(\pi)_{\text{red}}$, and put $L = \pi^*(-K_X)$. The standard resolution properties give a rational surface S and a simple-normal-crossing forest D of smooth rational curves. Its irreducible components are independent in $\text{Pic}(S)$ and number $\rho(S) - 1$. The connected components of this forest correspond exactly to the singular points of X .

A curve on S not contained in D will be called *exterior*. Suppose a counterexample to Theorem 1.1 exists over a fixed field, and choose one with $\rho(S)$ minimal. Choose an exterior (-1) -curve P for which $\ell = L \cdot P$ is least. An extremal-ray argument on the smooth surface proves that $L + \ell K_S$ is nef. Consequently every exterior integral curve Q satisfies

$$L \cdot Q \geq \ell(-K_S \cdot Q) \geq \ell.$$

This controls the exterior components of effective integral adjoint divisors, not only the (-1) -curves initially under consideration.

We use adjoint divisors of the form $K_S + A$, where A is an effective integral divisor formed from P and exceptional curves. Riemann–Roch produces effective representatives of these adjoint classes. Their small L -degree either forces a singular-point bound or produces a shorter exterior (-1) -curve. When P meets two disjoint exceptional (-2) -curves, they form a rational fiber with P of multiplicity two. A bisection is a curve whose morphism to the base of the ruling has degree two. The fiber intersection equations and the adjoint of a bisection then give the required bound. Certain remaining contact configurations admit birational replacements that do not decrease the number of singular points and strictly lower the Picard number of the minimal resolution. The replacements are realized by projective contractions; their anticanonical divisors and discrepancies are computed explicitly.

These arguments leave one possibility. There are pairwise disjoint exceptional curves C, B_1, B_2 of respective self-intersections $-2, -3, -\beta$, with $\beta \in \{3, 4, 5\}$, and an exterior (-1) -curve R disjoint from $C + B_1 + B_2 + P$, such that

$$R \sim K_S + C + B_1 + B_2 + 2P.$$

This identity bounds the weights and the number of edges of the entire exceptional forest. A direct classification gives ten possibilities. The full-rank lattice generated by D and P must have square absolute determinant, since the Picard lattice of a rational surface is unimodular. Seven possibilities fail this test. Riemann–Roch parity excludes two more; the last forces four isolated exceptional (-2) -curves whose sum is twice a Picard class. A separable double cover branched over such a sum contradicts the Hodge index theorem.

The bound is sharp in characteristic three. In every odd positive characteristic, a degree-two map from a rational bisection is separable and tame, and the cover branched over an even set of isolated exceptional nodes is separable. All other characteristic-dependent inputs used here hold in every positive characteristic. The examples in section 10 exhibit the failure of both degree-two mechanisms in characteristic two. Since the contractions use Keel’s positive-characteristic semiampness theorem, this argument is not asserted to extend to characteristic zero.

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2. RESOLUTIONS, INTERSECTION THEORY, AND PROJECTIVE CONTRACTIONS

2.1. Conventions. Unless a different field is specified, k is algebraically closed of positive characteristic. A surface is integral and projective over k . An integral curve is reduced and irreducible. On a smooth surface a $(-b)$ -curve means a smooth rational curve of self-intersection $-b$. We write $\sim$, $\sim_{\mathbb{Q}}$, and $\equiv$ for linear, rational linear, and numerical equivalence, respectively. A rational divisor is nef if it has nonnegative intersection with every integral curve, and it is semiample if a positive integral multiple is basepoint free. A nef divisor on a smooth surface is big precisely when its square is positive. Its null locus is the reduced union of curves of degree zero. We write $N^1(S)_{\mathbb{Q}}$ for rational divisor classes modulo numerical equivalence, and $\overline{\text{NE}}(S)$ for the closed cone of effective curve classes.

For a minimal resolution $\pi : S \rightarrow X$ of a klt del Pezzo surface of Picard number one, set

$$D = \text{Exc}(\pi)_{\text{red}} = \sum_i D_i, \quad L = \pi^*(-K_X), \quad v = L^2 = K_X^2.$$

This triple will be called a *resolution datum*. An exterior curve is an integral curve on S not contained in D . The canonical pullback will be written

$$K_S + \sum_i \lambda_i D_i = \pi^* K_X = -L.$$

Thus the discrepancy of D_i is $-\lambda_i$, and its log discrepancy is $1 - \lambda_i$. We prove below that $0 \leq \lambda_i < 1$. A coefficient attached to a (-2) -curve need not vanish when that curve is in a connected exceptional divisor containing a curve of smaller self-intersection. We therefore distinguish self-intersection from discrepancy throughout.

For a reduced simple-normal-crossing divisor $G = \sum G_i$, its dual graph has one vertex for each component and an edge for each intersection point. Simple normal crossings means that all components are smooth, every intersection is transverse, and no three components pass through one point. A *forest* has no cycles; its connected components are trees. We give its vertices weights $b_i = -G_i^2$ and write

$$A_G = -(G_i \cdot G_j), \quad q_i = K_S \cdot G_i = b_i - 2$$

when the components are rational. We call a vertex *weight-two* when $b_i = 2$ and *higher-weight* when $b_i \geq 3$. Contact vectors use the intersection numbers with the individual integral components, including their local intersection multiplicities. The notation $[b_1, \dots, b_m]$ denotes a chain with those weights; a bracket containing a curve name specifies that vertex rather than its weight.

We use the Hodge index theorem, adjunction, Riemann–Roch and Serre duality on smooth projective surfaces, as well as contraction of (-1) -curves and factorization of birational morphisms between smooth surfaces into point blowups. Our use of the surface minimal model program is over $\text{Spec } k$ and with zero boundary on a smooth surface; the needed contraction and termination results are [16, Theorems 4.4 and 1.1]. We also use the classical ruledness criterion in positive characteristic: a smooth projective surface of Kodaira dimension $-\infty$ is birationally ruled, with rational surfaces included. In particular a smooth surface with non-pseudo-effective canonical divisor has this property: it has $h^0(S, 12K_S) = 0$, so one may apply [4, Theorem 2.1(I)]. We write $p_g(S) = h^0(S, K_S)$ and $q(S) = h^1(S, \mathcal{O}_S)$; here q is the

arithmetic irregularity, which agrees with the genus of the base on a birationally ruled surface. For an integral curve C on a smooth surface, $p_a(C) = 1 + (C^2 + K_S \cdot C)/2$ denotes its arithmetic genus.

2.2. Inverse intersection matrices and rational trees. We recall two standard facts used in the contraction arguments.

Lemma 2.1 (Positivity of a Stieltjes inverse). *Let A be a symmetric positive-definite matrix with nonpositive off-diagonal entries. Then A^{-1} is entrywise nonnegative. If the graph of the nonzero off-diagonal entries is connected, then A^{-1} is entrywise positive.*

The first assertion is the usual inverse-positivity property of a positive-definite Stieltjes matrix; the strict assertion is its irreducible case.

Lemma 2.2 (Picard group of a rational tree). *Let Z be a connected reduced nodal curve whose irreducible components are copies of $\mathbb{P}^1$ and whose dual graph is a tree. The multidegree map*

$$\mathrm{Pic}(Z) \longrightarrow \mathbb{Z}^{\{\text{components of } Z\}}$$

is an isomorphism. In particular, a line bundle having degree zero on every irreducible component of Z is trivial.

This is the standard normalization description of the Picard group of a nodal curve: the gluing factor is $(k^*)^{b_1}$, where b_1 is the first Betti number of the dual graph, and is trivial for a tree.

2.3. Semiample and the minimal resolution.

Theorem 2.3 (Keel [6, Theorem 0.2]). *Let M be a nef line bundle on a scheme projective over a field of positive characteristic. Let $E(M)$ be the reduced union of the positive-dimensional subvarieties on which M is not big. Then M is semiample if and only if $M|_{E(M)}$ is semiample.*

The following collects standard resolution properties. Rationality is [2, Lemma 5.1]. We give the discrepancy and rank arguments in the form needed for the subsequent contractions, without invoking a classification of quotient singularities.

Proposition 2.4 (Geometry of the minimal resolution). *Let X be a rank-one klt del Pezzo surface over an algebraically closed field of positive characteristic, and let $\pi : S \rightarrow X$ be its minimal resolution. Put $D = \mathrm{Exc}(\pi)_{\mathrm{red}}$ and $L = \pi^*(-K_X)$. Then D is an SNC forest of smooth rational curves of square at most -2 . Its negative intersection matrix A is positive definite, and*

$$K_S + \sum_i \lambda_i D_i = \pi^* K_X, \quad \lambda = A^{-1} q, \quad q_i = -2 - D_i^2, \quad 0 \leq \lambda_i < 1.$$

The surface S is rational, $\mathrm{Pic}(S)$ is torsion-free and unimodular, $K_S^2 + \rho(S) = 10$, and $\#\mathrm{Irr}(D) = \rho(S) - 1$. Moreover $\#\pi_0(D) = n(X)$. In particular, S admits a birational morphism to $\mathbb{P}^2$ or a rational ruling.

Proof. The resolution is an isomorphism over the smooth locus of X and has connected fibers. Its exceptional connected components therefore correspond exactly to the singular points of X . First the exceptional classes are independent. A numerical relation, after separating signs, would give numerically equivalent effective exceptional divisors Z_+, Z_- with no common component. If both are nonzero then $Z_+^2 = Z_+ \cdot Z_- \geq 0$, contradicting Hodge index since $L \cdot Z_+ = 0$ and $L^2 > 0$. A relation of only one sign is excluded by an ample divisor. Hodge index now makes their intersection matrix negative definite. For an exceptional integral curve E put $b = -E^2 > 0$. Minimality excludes an exceptional smooth rational (-1) -curve; adjunction therefore gives $K_S \cdot E = 2p_a(E) - 2 + b \geq 0$. The inverse of the positive matrix with nonpositive off-diagonal entries is nonnegative by Lemma 2.1. Consequently the canonical pullback equations give $\lambda_i \geq 0$; klt gives $\lambda_i < 1$. If $p_a(E) \geq 1$, its equation $b\lambda_E = K_S \cdot E + \sum_{E' \neq E} (E \cdot E')\lambda_{E'}$ would give $\lambda_E \geq 1$. Thus every E has arithmetic genus

zero, is smooth rational, and has square at most -2 . If the incidence multigraph had a cycle, take its vertices J . Each vertex has total intersection at least two within J , so $A_J \mathbf{1} \leq q_J$. Restriction of the full discrepancy equations and positivity of A_J^{-1} would give $\lambda_J \geq A_J^{-1} q_J \geq \mathbf{1}$, a contradiction. This also excludes two intersection points or tangency between a pair. A triple intersection would give a triangle. Thus the exceptional divisor is an SNC forest.

Rationality follows from [2, Lemma 5.1]: the surface X is of del Pezzo type (take boundary zero), and its minimal resolution is birational to it. The standard blowup formulas on a rational surface give the Picard and canonical-square assertions, as well as a birational morphism to $\mathbb{P}^2$ or a Hirzebruch surface.

It remains to determine the rank of the exceptional divisor. For any ample integral H on S , put $a_i = H \cdot D_i$, $x = A^{-1}a$, and $M = H + \sum_i x_i D_i$. Then M is nef, $M^2 = H^2 + a^\top A^{-1}a > 0$, and its exact null locus is D . A divisible multiple restricts trivially to D by Lemma 2.2; Theorem 2.3 makes it semiample. Its Stein contraction contracts precisely the same connected curves as π . The image of the common graph has singleton fibers over each normal target; the projections are proper birational and quasi-finite, hence are isomorphisms. Thus its target is X . Hence $[M] \in \pi^* N^1(X)_\mathbb{Q}$. The map $[H] \mapsto [M]$ is the linear orthogonal projection onto $\langle D \rangle^\perp$. Ample classes span $N^1(S)_\mathbb{Q}$, so its image has dimension at most $\rho(X) = 1$. It contains $L \neq 0$. The independent exceptional curves therefore number $\rho(S) - 1$. $\square$

Lemma 2.5 (Square and degree formulas). *Let $B = \sum_i \lambda_i G_i$ with $A\lambda = q$, and put*

$$H = -(K_T + B).$$

Then

$$H^2 = K_T^2 + q^\top A^{-1}q. \quad (2.1)$$

If $C \not\subset G$ is an irreducible curve and $p_i = C \cdot G_i$, then

$$H \cdot C = -K_T \cdot C - p^\top \lambda. \quad (2.2)$$

In particular, if C is a (-1) -curve, then

$$H \cdot C = 1 - p^\top \lambda. \quad (2.3)$$

Proof. The relation $A\lambda = q$ gives

$$B^2 = -\lambda^\top A\lambda = -\lambda^\top q, \quad K_T \cdot B = \lambda^\top q.$$

This proves (2.1). Formula (2.2) is the definition of H , and adjunction gives $-K_T \cdot C = 1$ for a (-1) -curve. $\square$

2.4. Anticanonical contractions.

Theorem 2.6 (Anticanonical contraction to a rank-one klt del Pezzo surface). *Let T be a smooth projective surface over an algebraically closed field of positive characteristic. Let*

$$G = \sum_{i=1}^r G_i$$

be a reduced simple-normal-crossing forest of smooth rational curves. Assume that the classes $[G_i]$ are linearly independent and that

$$r = \rho(T) - 1.$$

Let

$$B = \sum_{i=1}^r \lambda_i G_i, \quad \lambda_i < 1, \quad H = -(K_T + B),$$

where the λ_i are rational. Suppose that

- (i) *H is nef and big;*
- (ii) *an irreducible curve $C \subset T$ satisfies $H \cdot C = 0$ if and only if C is one of the G_i .*

Then a sufficiently divisible multiple of H defines a projective birational morphism

$$f : T \longrightarrow Y$$

with connected fibers and exceptional locus G . The target Y is a normal projective rank-one klt del Pezzo surface, and, after choosing compatible canonical divisors,

$$K_T + B = f^*K_Y, \quad -K_Y \text{ is ample}, \quad \rho(Y) = 1. \quad (2.4)$$

Proof. Choose a positive integer m such that mH and mB are integral. Since H is nef and big on a surface, the exceptional locus $E(mH)$ in Theorem 2.3 is precisely the reduced union of curves C with $H \cdot C = 0$, hence it is G . The line bundle $\mathcal{O}_T(mH)$ has degree zero on every component of every connected component of G . By Lemma 2.2, its restriction to each such component is trivial. Keel's theorem therefore shows that mH is semiample.

Let $f : T \rightarrow Y$ be the Stein factorization of the morphism defined by a sufficiently divisible basepoint-free multiple of mH . Since H is big, Y is a surface. The equality $f_*\mathcal{O}_T = \mathcal{O}_Y$ implies that the induced extension of function fields is trivial, so f is birational. A curve is contracted by f exactly when it has degree zero against H ; hence $\text{Exc}(f) = G$. There is an ample Cartier divisor A on Y and a positive integer n such that

$$nmH \sim f^*A. \quad (2.5)$$

Indeed, after Stein factorization A is the pullback of the hyperplane class from the image, and a finite pullback of an ample divisor is ample.

The pullback of $N^1(Y)_{\mathbb{Q}}$ is injective and orthogonal to all $r = \rho(T) - 1$ independent exceptional curves, so $\rho(Y) \leq 1$. Since Y is projective, it has an ample class and $\rho(Y) \geq 1$. Therefore $\rho(Y) = 1$.

It remains to identify the canonical divisor on Y . Choose a nonzero rational two-form on the common function field of T and Y , and let K_T and K_Y be its divisors on the two surfaces. Then $f_*K_T = K_Y$. From (2.5),

$$K_T + B = -H \sim_{\mathbb{Q}} f^*\left(-\frac{1}{nm}A\right).$$

Pushing forward gives

$$K_Y \sim_{\mathbb{Q}} -\frac{1}{nm}A.$$

Thus K_Y is $\mathbb{Q}$ -Cartier and $-K_Y$ is ample. Put

$$E = K_T + B - f^*K_Y.$$

The divisor E is exceptional and $\mathbb{Q}$ -linearly trivial. The intersection matrix of the exceptional curves is negative definite by the Hodge index theorem, so an exceptional numerically trivial divisor is zero. Hence $E = 0$, proving the crepant equality in (2.4).

Finally, the simple-normal-crossing sub-pair (T, B) is sub-klt because every coefficient of B is less than one. Indeed, blowing up a point on components with coefficients b_1, b_2 gives exceptional coefficient $b_1 + b_2 - 1 < 1$; at a point on just one component it gives $b_1 - 1 < 1$, and away from the boundary it gives -1 . Repeated point blowups therefore keep all coefficients below one. The computation is unchanged when some coefficients are negative. The crepant equality $K_T + B = f^*K_Y$ identifies the discrepancies of $(Y, 0)$ with those of (T, B) on every higher resolution. Hence Y is klt; compare the discrepancy formalism in [8, Section 2.3]. $\square$

Theorem 2.7 (Numerical rank-one contraction on a rational surface). *Let T be a smooth projective rational surface over k , and let*

$$G = \sum_{i=1}^r G_i$$

be a reduced simple-normal-crossing forest of smooth rational curves whose intersection matrix is negative definite. Assume that the classes of the G_i are independent and that $r = \rho(T) - 1$.

Let

$$B = \sum_i \beta_i G_i, \quad \beta_i < 1, \quad H = -(K_T + B),$$

where the coefficients are rational. Suppose that

$$H \cdot G_i = 0 \quad \text{for every } i, \quad H^2 > 0,$$

and that $H \cdot C_0 > 0$ for one irreducible curve $C_0 \not\subset G$. Then G has a projective contraction

$$f : T \longrightarrow Y$$

to a normal rank-one klt del Pezzo surface, and

$$K_T + B = f^* K_Y.$$

Proof. Choose an ample integral divisor A_0 on T , put $a_i = A_0 \cdot G_i$, and let $x = A_G^{-1}a$. By Lemma 2.1, $x_i > 0$. The rational divisor

$$M = A_0 + \sum_i x_i G_i$$

is orthogonal to G , has square $A_0^2 + a^\top A_G^{-1}a > 0$, and has positive degree on every integral curve outside G . Thus M is nef and big with null locus exactly G . Both H and M lie in the one-dimensional orthogonal complement of the independent classes of G . Write $H \equiv cM$. The hypothesis $H \cdot C_0 > 0$, together with $M \cdot C_0 > 0$, gives $c > 0$. Consequently H itself is nef and big with null locus exactly G . The conclusion, including the descent of the canonical divisor, now follows from Theorem 2.6. $\square$

2.5. Orthogonal projection and nonexceptional curves. For the following identity write $f : T \rightarrow Y$ for a minimal rank-one klt del Pezzo resolution, $G = \text{Exc}(f)_{\text{red}}$, $A = A_G$, and $L = f^*(-K_Y)$.

Lemma 2.8 (Rank-one projection identity). *Let $P \not\subset G$ be a (-1) -curve, and let $p_i = P \cdot G_i$. Then*

$$p^\top A^{-1}p = 1 + \frac{(L \cdot P)^2}{K_Y^2} > 1. \quad (2.6)$$

Moreover,

$$p^\top A^{-1}q = p^\top \lambda = 1 - L \cdot P < 1. \quad (2.7)$$

Proof. Put $u = A^{-1}p$ and

$$\bar{P} = P + \sum_i u_i G_i.$$

Then $\bar{P}$ is orthogonal to every component of G . Since $\rho(Y) = 1$, the orthogonal complement of the exceptional span in $N^1(T)_{\mathbb{Q}}$ is generated by L . Hence

$$\bar{P} \equiv \frac{L \cdot P}{K_Y^2} L.$$

On the other hand,

$$\bar{P}^2 = -1 + p^\top A^{-1}p.$$

Comparing squares proves (2.6). Formula (2.7) follows from (2.3). $\square$

Lemma 2.9 (Null curves outside the minimal exceptional divisor). *Let (T, D, L) be the resolution datum of a klt del Pezzo surface of Picard number one, and let G be a nef and big divisor on T . If an irreducible curve $R \not\subset D$ satisfies $G \cdot R = 0$, then*

$$R^2 = K_T \cdot R = -1, \quad p_a(R) = 0.$$

There are only finitely many such curves, and any two of them are disjoint.

Proof. The Hodge index theorem gives $R^2 < 0$. Since $R \not\subset D$,

$$K_T \cdot R = -L \cdot R - \sum_{C \subset D} \lambda_C(C \cdot R) < 0.$$

Adjunction gives

$$2p_a(R) - 2 = R^2 + K_T \cdot R.$$

The right side is a sum of two negative integers, while the left side is at least -2 . Hence both integers are -1 and $p_a(R) = 0$. The lattice $G^\perp \cap \text{NS}(T)$ is negative definite, so it contains only finitely many integral vectors of square -1 . Finally, if R_1 and R_2 were two such null curves with $R_1 \cdot R_2 > 0$, the matrix

$$\begin{pmatrix} -1 & R_1 \cdot R_2 \\ R_1 \cdot R_2 & -1 \end{pmatrix}$$

would fail to be negative definite. Thus $R_1 \cdot R_2 = 0$. $\square$

3. RATIONAL PENCILS AND NEF THRESHOLDS

A *rational ruling* is a morphism $f : S \rightarrow \mathbb{P}^1$ with connected fibers and smooth rational geometric generic fiber. Its fiber class is the divisor class of a scheme-theoretic fiber. A class in the free Picard group of a rational surface is *primitive* if it is not an integer multiple dG with $d > 1$.

3.1. Nef divisors of square zero and rational pencils.

Lemma 3.1 (A nef divisor of square zero and canonical degree minus two). *Let F be an integral nef divisor on a smooth projective rational surface T . If $F^2 = 0$ and $K_T \cdot F = -2$, then F is primitive and $|F|$ is a basepoint-free complete pencil defining a generically separable rational ruling with connected fibers.*

Proof. If $F = dG$ in the torsion-free Picard lattice with $d > 1$, then $d = 2$ and $G^2 = 0$, $K_T \cdot G = -1$, contradicting the parity $G^2 - K_T \cdot G \in 2\mathbb{Z}$ supplied by Riemann–Roch. Thus F is primitive. Since $(K_T - F) \cdot F = -2$, Serre duality and Riemann–Roch give $h^0(F) \geq 2$. Write $|F| = Z + |M|$ with fixed divisor Z and nonzero moving divisor M . A linear system without fixed components has nef moving class. Both M and Z are F -null. Hodge index on $F^\perp$ gives $M^2 \leq 0$, whereas nefness gives $M^2 \geq 0$; its equality case gives $M \equiv cF$ with $c > 0$. Primitivity and integrality imply $c \in \mathbb{Z}_{>0}$. Intersection with an ample divisor and $Z \geq 0$ give $c \leq 1$, so $c = 1$ and $Z = 0$. Two suitably chosen members now have no common component and zero intersection, so there is no base point. The image is a curve since $F^2 = 0$ and the system is nonconstant. In the Stein factorization $f : T \rightarrow B$, the normal projective base is smooth and $H^1(B, \mathcal{O}_B)$ injects into $H^1(T, \mathcal{O}_T) = 0$. Hence $B \simeq \mathbb{P}^1$. If G_0 is the class of a fiber, then $F \sim dG_0$, where d is the positive degree of the line bundle on B defining the complete system. Primitivity forces $d = 1$; $f_*\mathcal{O}_T = \mathcal{O}_B$ then gives $h^0(F) = 2$. The Stein condition makes $k(B)$ relatively algebraically closed in $k(T)$. A parameter of $k(B)$ cannot be a p -th power in $k(T)$, so its differential is nonzero. The field extension is separably generated and regular. Thus the geometric generic fiber is integral. Flatness over the smooth base and adjunction give its arithmetic genus zero; it is therefore a smooth rational curve. This proves the ruling assertion. $\square$

Lemma 3.2 (Primitive rational rulings and their fibers). *Let $f : S \rightarrow \mathbb{P}^1$ be a rational ruling on a smooth projective rational surface with primitive fiber class F . Every irreducible fiber is a reduced smooth rational curve. There is a factorization of f by smooth vertical (-1) -curve contractions to a Hirzebruch surface. It can start with any prescribed vertical (-1) -curve. Every reduced fiber of f is an SNC tree of smooth rational curves, and a vertical (-1) -curve has valency at most two. If n_t is the number of reduced components of the reducible fiber at t , then*

$$\rho(S) = 2 + \sum_t (n_t - 1). \quad (3.1)$$

Proof. The morphism is flat, since its base is a smooth curve and S is smooth; every scheme-theoretic fiber has arithmetic genus zero. If a fiber is mC with irreducible reduced support, primitivity gives $m = 1$, and arithmetic genus zero makes C smooth rational. For a reducible connected fiber $\sum \mu_i C_i$, its intersection with each C_j gives $C_j^2 < 0$. The equality $-2 = K_S \cdot F = \sum \mu_i (K_S \cdot C_i)$ selects a component of negative canonical degree. Adjunction makes it a smooth rational (-1) -curve. Contract this curve, or start with any prescribed vertical (-1) -curve. The map descends because it is constant on the exceptional curve; the descended fiber has primitive class, since its pullback is F . Iterating lowers the Picard number and ends with all fibers smooth $\mathbb{P}^1$'s. The resulting smooth $\mathbb{P}^1$ -fibration is a projective bundle: $\mathrm{Br}(\mathbb{P}^1) = 0$ over an algebraically closed field. Splitting its rank-two bundle gives a Hirzebruch surface of Picard number two. Reversing the contractions blows up points of smooth fibers or their nodes; this preserves SNC fiber trees and proves the valency assertion, including for the prescribed first contraction. Every blowup increases both ρ and $\sum_t (n_t - 1)$ by one. This proves the formula. $\square$

3.2. The nef threshold of the anticanonical pullback. All degrees in this subsection are measured against $L = \pi^*(-K_X)$, not against $-K_S$. The next lemma is the nef-threshold argument specialized to the smooth resolution; its contraction input is Tanaka's theorem. Whenever exterior (-1) -curves exist, write

$$\ell_{\mathrm{ext}} = \min\{L \cdot P : P \text{ is an exterior } (-1)\text{-curve}\}.$$

The minimum exists because a Cartier multiple of L has positive integral degree on every such curve. In the next lemma, $\ell = \ell_{\mathrm{ext}}$.

Lemma 3.3 (The least exterior degree is the nef anticanonical threshold). *Let (S, D, L) be the resolution datum of a klt del Pezzo surface of Picard number one in positive characteristic with $\rho(S) > 2$. Exterior (-1) -curves exist, and their least positive L -degree ℓ satisfies*

$$N := L + \ell K_S \text{ is nef}, \quad \sup\{t \geq 0 : L + t K_S \text{ is nef}\} = \ell.$$

Every exterior integral curve C satisfies the stronger inequality

$$L \cdot C \geq \ell(-K_S \cdot C) \geq \ell. \quad (3.2)$$

In particular an effective integral divisor of L -degree less than ℓ is supported on D and has L -degree zero. In addition,

$$N \cdot D_i = \ell(-2 - D_i^2), \quad N^2 = (1 - 2\ell)L^2 + \ell^2 K_S^2 \geq 0.$$

Proof. Apply [16, Theorem 4.4] over $\mathrm{Spec} k$ to the smooth projective surface S with boundary $\Delta = 0$. The base field is excellent, S is $\mathbb{Q}$ -factorial, and K_S is Cartier, so all its hypotheses hold. Every K_S -negative extremal ray contracts by a projective morphism $f : S \rightarrow Y$ with $f_* \mathcal{O}_S = \mathcal{O}_Y$ and $\rho(Y) = \rho(S) - 1 > 1$. Thus $\dim Y = 2$: a projective integral curve has Picard number one, and a point has Picard number zero. The morphism is birational and has an exceptional curve E . Then $E^2 < 0$ and $K_S \cdot E < 0$, so adjunction makes E a smooth rational (-1) -curve generating that ray. It is exterior because no component of the minimal exceptional divisor has square -1 . Such a ray exists since $K_S \cdot L = -L^2 < 0$ makes K_S nonnef. Here one uses the standard compact ample slice of the closed curve cone to detect nonnefness on an extremal ray. A Cartier multiple of L makes the positive degrees of all exterior (-1) -curves discrete, so their minimum $\ell > 0$ exists. If $N = L + \ell K_S$ were nonnef, choose an N -negative extremal ray. Since L is nef, that ray is K_S -negative and hence generated by an exterior (-1) -curve E . But $N \cdot E = L \cdot E - \ell \geq 0$, a contradiction. For $0 \leq t \leq \ell$, $L + t K_S$ is a convex combination of L and N ; for $t > \ell$ it is negative on a curve attaining the minimum. This proves the exact threshold. For an exterior integral curve,

$$K_S \cdot C = -L \cdot C - \sum_i \lambda_i (D_i \cdot C) < 0$$

is an integer, so nefness of N proves (3.2). The effective-divisor assertion follows by summing components. The final identities are direct intersections, using $L \cdot D_i = 0$ and $K_S \cdot L = -L^2$. $\square$

4. BIRATIONAL REPLACEMENTS AND SINGULAR-POINT COUNTS

When an exterior (-1) -curve meets the exceptional divisor normally in at most two points, one of them on a unique (-2) -curve, removing that (-2) -curve and blowing down the exterior curve is the elementary swap of [13, Definition 1.3]. Its preservation properties are proved in [13, Lemma 2.9]. We use the same geometric operation in a wider numerical form, with explicit anticanonical coefficients and exact singular-point counts.

Fix a resolution datum (S, D, L) and an exterior (-1) -curve P . Put $p_i = P \cdot D_i$ and $q_i = K_S \cdot D_i = -2 - D_i^2$. A contact at D_i is called *excess* if $p_i > q_i$ and *bounded* if $p_i \leq q_i$. These terms refer only to the displayed inequality, not to a transversality assumption. We do not assume that $D + P$ is a simple-normal-crossing divisor. A replacement theorem below will explicitly require its retained divisor to be a simple-normal-crossing forest.

4.1. Restrictions on intersection multiplicities.

Lemma 4.1 (Elementary discrepancy bound). *Let $D_i^2 = -b$ with $b \geq 3$. Then its discrepancy coefficient satisfies*

$$\lambda_i \geq \frac{b-2}{b}.$$

Proof. Write the positive intersection matrix with D_i first:

$$A = \begin{pmatrix} b & -v^\top \\ -v & M \end{pmatrix}.$$

The Schur complement at D_i is at most b , so

$$(A^{-1})_{ii} \geq \frac{1}{b}.$$

Since the canonical-degree vector satisfies $q \geq (b-2)e_i$ and A^{-1} is entrywise nonnegative,

$$\lambda_i = e_i^\top A^{-1} q \geq (b-2)(A^{-1})_{ii} \geq \frac{b-2}{b}.$$

□

Lemma 4.2 (Existence and rigidity of an excess contact). *The contact vector p has at least one excess contact. Every nonzero bounded contact is simple. If a higher-weight component $D_i^2 = -b$, $b \geq 3$, has an excess contact, then*

$$b = 3, \quad P \cdot D_i = 2.$$

In particular, there is at most one higher-weight excess contact.

Proof. If $p \leq q$ coordinatewise, then positivity of A^{-1} and Lemma 2.8 give

$$p^\top A^{-1} p \leq p^\top A^{-1} q = p^\top \lambda = 1 - L \cdot P < 1,$$

contrary to the projection identity.

Suppose that $0 < p_i \leq q_i$ and $p_i \geq 2$. Then $q_i \geq p_i \geq 2$, and Lemma 4.1 gives

$$p_i \lambda_i \geq \frac{p_i q_i}{q_i + 2} \geq 1,$$

again contradicting $p^\top \lambda < 1$. Thus every bounded contact is simple.

Finally, if $D_i^2 = -b$ is a higher-weight excess contact, then $p_i \geq b-1$. Hence

$$1 > p_i \lambda_i \geq \frac{(b-1)(b-2)}{b}.$$

This forces $b = 3$. Since then $\lambda_i \geq 1/3$, the inequality $p_i \lambda_i < 1$ excludes $p_i \geq 3$; hence $p_i = 2$. Two such excess contacts would contribute at least $4/3$ to the total discrepancy sum, so at most one exists. □

Lemma 4.3 (Discrepancy separation in one exceptional component). *Let B_0 and B_n be distinct higher-weight vertices in one connected component of D . Then*

$$\lambda_{B_0} + \lambda_{B_n} \geq 1.$$

Proof. Let $B_0 = v_0 - \cdots - v_n = B_n$ be the path joining the two vertices and let A_P be its positive intersection matrix, with the original diagonal weights. Restricting the full discrepancy equations to the path gives

$$A_P \lambda_P = q_P + c, \quad c \geq 0.$$

Since A_P^{-1} is entrywise nonnegative, $\lambda_P \geq \theta := A_P^{-1} q_P$. Put $u = \mathbf{1} - \theta$. For a path,

$$A_P u = e_0 + e_n.$$

Let A_0 be the path matrix with endpoint weights 3, 3 and every interior weight 2. Then $A_P \succeq A_0$, and inversion reverses the Loewner order. Since

$$A_0 \left(\frac{1}{2} \mathbf{1} \right) = e_0 + e_n,$$

we obtain

$$u_0 + u_n = (e_0 + e_n)^\top A_P^{-1} (e_0 + e_n) \leq 1.$$

Therefore $\theta_0 + \theta_n \geq 1$, and the same holds for λ . $\square$

Lemma 4.4 (Valencies of exceptional trees). *Every vertex of the dual graph of D has valency at most three.*

Proof. At a vertex of weight b , let the d neighbours have weights b_j . They are pairwise nonadjacent. The principal star is negative definite, so $u = \sum_j 1/b_j < b$. Its discrepancy inequalities give

$$b\lambda \geq b - 2 + \sum_j \lambda_j, \quad b_j \lambda_j \geq b_j - 2 + \lambda, \quad (b - u)\lambda \geq b + d - 2 - 2u.$$

Since $\lambda < 1$, one obtains $u > d - 2$. But $u \leq d/2$, so $d < 4$. $\square$

4.2. Replacing one exceptional component. Choose a component $C \subset D$ and write $D = C + D_0$. Order the positive intersection matrix, canonical-degree vector, and contact vector as

$$A = \begin{pmatrix} b & -v^\top \\ -v & M \end{pmatrix}, \quad q = \begin{pmatrix} b-2 \\ q_0 \end{pmatrix}, \quad p = \begin{pmatrix} m \\ r \end{pmatrix}, \quad (4.1)$$

where $C^2 = -b$ and $m = P \cdot C$. Assume that every contact with D_0 is bounded. By Lemma 4.2,

$$r \in \{0, 1\}^{D_0}, \quad q_0 \geq r. \quad (4.2)$$

Define

$$g = (b - v^\top M^{-1} v)^{-1}, \quad \theta = M^{-1} q_0, \quad u = M^{-1} r, \quad (4.3)$$

$$\mu = g(b - 2 + v^\top \theta), \quad h = r^\top u, \quad \eta = r^\top \theta, \quad (4.4)$$

$$\alpha = v^\top u, \quad \tau = m + \alpha, \quad \ell = 1 - \eta - \tau\mu = L \cdot P. \quad (4.5)$$

Here μ is the original discrepancy coefficient of C , while the original discrepancy vector on D_0 is

$$\lambda_0 = \theta + \mu M^{-1} v. \quad (4.6)$$

Theorem 4.5 (Replacement of one exceptional component). *Assume that C is the only excess contact of P and that*

$$G^\dagger = P + D_0$$

is a simple-normal-crossing forest. Then

$$0 \leq h \leq \eta < 1.$$

Put

$$c = \frac{1 - \eta}{1 - h}, \quad \lambda_0^\dagger = \theta - cu = (M - rr^\top)^{-1}(q_0 - r). \quad (4.7)$$

The divisor

$$L^\dagger = -K_S + cP - \sum_{D_i \subset D_0} \lambda_i^\dagger D_i \quad (4.8)$$

is nef and big and has exact null locus $G^\dagger$. A sufficiently divisible multiple contracts $G^\dagger$ to a rank-one klt del Pezzo surface $X^\dagger$. On the nonminimal resolution $S \rightarrow X^\dagger$ the coefficient of P in the boundary convention of Theorem 2.6 is $-c$, while the coefficients on D_0 are $\lambda_0^\dagger$. They satisfy

$$-c < 1, \quad 0 \leq \lambda_0^\dagger \leq \lambda_0 < 1. \quad (4.9)$$

Moreover,

$$(L^\dagger)^2 - L^2 = \frac{(1-\eta)^2}{1-h} - \frac{\mu^2}{g} > 0, \quad (4.10)$$

$$L^\dagger \cdot C = \tau \frac{1-\eta}{1-h} - \frac{\mu}{g} > 0. \quad (4.11)$$

Let $\sigma : S \rightarrow S_1$ be the blowdown of P . The induced morphism $S_1 \rightarrow X^\dagger$ is the minimal resolution, with exceptional divisor the image of D_0 . In particular

$$\rho(S_1) = \rho(S) - 1.$$

If Γ is the connected component of D containing C , put

$$s_C = \#\pi_0(\Gamma - C),$$

and let a be the number of connected components of $D - C$ met by the remaining contact vector r . Then

$$\#\text{Sing}(X^\dagger) - \#\text{Sing}(X) = \begin{cases} s_C - a, & r \neq 0, \\ s_C - 1, & r = 0. \end{cases} \quad (4.12)$$

Proof. The bounded-contact inequalities and Lemma 4.2 give $q_0 \geq r$, $r_i \in \{0, 1\}$ and $0 \leq h \leq \eta \leq r^\top \lambda_0 < 1$. Thus $M - rr^\top$ is positive definite with nonpositive off-diagonal entries. The rank-one inverse formula proves (4.7); Stieltjes positivity gives $0 \leq \lambda_0^\dagger = \theta - cu \leq \theta \leq \lambda_0 < 1$. The negative intersection matrix and canonical-degree vector of the retained forest are

$$A^\dagger = \begin{pmatrix} 1 & -r^\top \\ -r & M \end{pmatrix}, \quad q^\dagger = \begin{pmatrix} -1 \\ q_0 \end{pmatrix}.$$

Its Schur complement is positive definite, and direct multiplication sends $(-c, \lambda_0^\dagger)^\top$ to $q^\dagger$. This proves the null equations and independence of all retained classes. On the original support, block inversion gives

$$p^\top A^{-1} p = h + g\tau^2 > 1, \quad p^\top \lambda = \eta + \tau\mu = 1 - \ell. \quad (4.13)$$

The respective correction terms after and before the replacement are $q_0^\top M^{-1} q_0 + (1-\eta)^2/(1-h)$ and $q_0^\top M^{-1} q_0 + \mu^2/g$. Their difference is (4.10); after multiplication by $g(1-h) > 0$ its numerator is

$$g\ell^2 + 2g\tau\mu\ell + \mu^2(h + g\tau^2 - 1) > 0.$$

The deleted degree equals $\tau c - \mu/g$; its numerator after the same multiplication is $g\tau\ell + \mu(h + g\tau^2 - 1) > 0$. Here $\tau \geq m > 0$, so strictness includes $\mu = 0$. Finally

$$L^\dagger = L + \mu C + \sum_{D_i \subset D_0} (\lambda_i - \lambda_i^\dagger) D_i + cP. \quad (4.14)$$

Every added coefficient is nonnegative. Curves outside $D \cup P$ therefore have positive degree, as does the deleted C ; the exact null locus is $P + D_0$. Its rank is $\rho(S) - 1$, its boundary coefficients are less than one, and its square is positive. The SNC forest hypothesis now permits Theorem 2.6, proving the claimed projective klt anticanonical contraction.

There are at most two retained contacts on P : each is simple and higher-weight, has coefficient at least $1/3$, and their total original discrepancy sum is less than one. Because $P + D_0$ is an SNC forest, these contacts are distinct points on P in different connected

components of D_0 , and neither is an exceptional node. Blowing down P therefore preserves smoothness of the retained curves and gives at most one transverse double point between them. The new negative intersection matrix and canonical-degree vector are $M - rr^\top$ and $q_0 - r$; all retained squares are at most -2 . The induced resolution $S_1 \rightarrow X^\dagger$ is consequently minimal, with $\rho(S_1) = \rho(S) - 1$. Deleting C replaces its exceptional connected component by s_C pieces. With no retained contact, the isolated contracted P gives a smooth point, so the singular count changes by $s_C - 1$. Otherwise its blowdown joins precisely the a contacted pieces, changing that count by $s_C - a$. Every surviving component contains only curves of square at most -2 and has a singular image; each therefore contributes a singular point. $\square$

4.3. Birational modifications involving an isolated (-2) -curve.

Theorem 4.6 (The isolated- A_1 exchange with its two-curve contraction). *Let W be an isolated original (-2) -curve and let the exterior (-1) -curve P meet W and one vertex B of another exceptional connected component Δ , once each, with no other original contact. If B has weight two, $W + 2P + B$ is a complete rational fiber. Otherwise, with $A = A_\Delta$, $q = q_\Delta$, $e = e_B$, put $h = e^\top A^{-1}e$ and $\eta = e^\top A^{-1}q$. Then*

$$B^2 = -3, \quad \frac{1}{2} < h \leq \eta < 1. \quad (4.15)$$

Write d for the valency of B in Δ ; it is positive. Blow down P and then the image of W , obtaining $f : S \rightarrow T$. The images of $D - W - B$ on T contract to a rank-one klt del Pezzo surface X_0 , with minimal resolution T , and

$$K_{X_0}^2 - K_X^2 = 2 - \eta^2/h > 0, \quad \# \text{Sing}(X_0) - \# \text{Sing}(X) = d - 2. \quad (4.16)$$

Thus for $d \geq 2$ the replacement does not decrease the number of singular points and lowers the minimal-resolution Picard number by two. If $d = 1$, let C be the unique neighbour of B . Blow up the transverse intersection of B_T and C_T on T . Retain the strict transform of B_T as an isolated A_1 , omit the new exceptional (-1) -curve, and retain the transformed $D - W - B$. The support contracts to a rank-one klt del Pezzo surface X_1 with unchanged singular-point count, minimal-resolution Picard decrease one, and

$$K_{X_1}^2 - K_X^2 = \frac{2(2\eta(1 - \eta) + 2h - 1)}{4h - 1} > 0. \quad (4.17)$$

Proof. The weight-two case follows from the support intersections and Lemma 3.1. In the other case Lemma 2.8 gives $1/2 + h > 1$, and the discrepancy sum gives $\eta < 1$. If $B^2 = -b$, then $\eta \geq (b - 2)h$, forcing $b = 3$ and $h \leq \eta$. An isolated B would have $h = 1/3$, so $d \geq 1$. Order B first, write

$$A = \begin{pmatrix} 3 & -a^\top \\ -a & A_0 \end{pmatrix}, \quad q = \begin{pmatrix} 1 \\ q_0 \end{pmatrix}, \quad \lambda_0 = A_0^{-1}q_0, \quad u = A_0^{-1}a, \quad s = a^\top \lambda_0, \quad t = a^\top u.$$

Schur complementation gives

$$s = \eta/h - 1, \quad t = 3 - 1/h, \quad q^\top A^{-1}q = q_0^\top \lambda_0 + \eta^2/h, \quad \lambda|_{\Delta-B} = \lambda_0 + \eta u. \quad (4.18)$$

In particular $0 \leq s < 1$, $1 < t < 2$, and $0 \leq \lambda_0 < 1$. The two given points of P are distinct and avoid exceptional nodes. After the first blowdown W has square -1 and B has square -2 ; after the second, B_T is a smooth (-1) -curve. The map f is an isomorphism near $D - W - B$, and

$$K_S = f^*K_T + W + 2P, \quad f^*B_T = B + W + 2P.$$

Thus the retained matrix on T is unchanged. Its anticanonical candidate has square $L^2 + 2 - \eta^2/h > 0$ and has degree

$$1 - s = 2 - \eta/h > 0$$

on the exterior curve B_T . Its boundary vector is λ_0 on $\Delta - B$, with the other exceptional components unchanged. The support is a negative-definite SNC forest of rank $\rho(T) - 1$ and all its squares are at most -2 . The criterion Theorem 2.7 gives X_0 and its minimal resolution.

Removing W loses one connected component and deleting B replaces Δ by d pieces, so the count is $d - 2$.

For $d = 1$ one has $a = e_C$. The point $B_T \cap C_T$ is the unchanged original B - C point, disjoint from the two contracted curves and from every other retained component. Its blowup separates the two curves: B becomes an isolated (-2) -curve, C loses one in its square, and the new exceptional E meets just these two strict transforms, once each at distinct points. The remaining negative intersection matrix and canonical-degree vector are $A_0 + e_C e_C^T$ and $q_0 + e_C$. Their solution is

$$\lambda_0^\dagger = \lambda_0 + \varepsilon u, \quad \varepsilon = \frac{1-s}{1+t} = \frac{2h-\eta}{4h-1}. \quad (4.19)$$

Here $0 < \varepsilon < \eta$, because $\eta - \varepsilon = 2h(2\eta - 1)/(4h - 1) > 0$. Thus every new coefficient is in $[0, 1)$, by comparison with the original $\lambda_0 + \eta u$. The new anticanonical candidate has positive degree $1 - (\lambda_0^\dagger)_C = \varepsilon$ on the exceptional curve E . The square change equals

$$1 + 2s + t - \frac{(s+t)^2}{1+t} - \frac{\eta^2}{h},$$

which is exactly (4.17). The criterion applies again. All retained squares are at most -2 , so this smooth surface is the minimal resolution. It has Picard number $\rho(S) - 1$, and its two local exceptional connected components are precisely the isolated new B and the connected nonempty $\Delta - B$. No other exceptional connected component is changed. $\square$

5. EVEN SETS OF EXCEPTIONAL CURVES AND THE PICARD LATTICE

An *isolated exceptional node* is a connected component of D consisting of one (-2) -curve. If $W_1, \dots, W_s$ are these components, put

$$\mathcal{C}_D = \ker \left(\mathbb{F}_2^s \longrightarrow \text{Pic}(S)/2\text{Pic}(S), \quad (\epsilon_i) \longmapsto \sum_i \epsilon_i [W_i] \right).$$

This is the standard code of disjoint nodal curves [5, Section 2], here restricted to the isolated components of the exceptional divisor. It records divisibility of their sums in $\text{Pic}(S)$, not the existence of arbitrary classes of square -2 . We will also use the standard parity fact: if a sum of w disjoint (-2) -curves is twice a Picard class, then $4 \mid w$. This follows from Riemann–Roch parity and holds in every characteristic; compare [5, Section 2]. The next argument adapts the classical branched-cover method to the full exceptional forest of a rank-one klt del Pezzo surface. Its rank comparison includes every unbranched exceptional component.

Theorem 5.1 (Vanishing of the isolated-node code). *Let X be a rank-one klt del Pezzo surface over an algebraically closed field of positive characteristic different from two, and let*

$$\pi : S \longrightarrow X, \quad D = \text{Exc}(\pi)_{\text{red}}, \quad L = \pi^*(-K_X)$$

be its minimal resolution. Assume that S is rational, that the connected components of D are trees of smooth rational curves, and that the irreducible components of D are linearly independent and number $\rho(S) - 1$. Then

$$\mathcal{C}_D = 0.$$

Equivalently, no nonempty sum of isolated nodes is twice a divisor in $\text{Pic}(S)$.

Proof. Suppose that a nonempty collection of isolated nodes satisfies

$$B = W_1 + \dots + W_n = 2M$$

in $\text{Pic}(S)$. The curves W_i are pairwise disjoint. Since the characteristic is not two, the double cover

$$g : Y \longrightarrow S$$

associated with M and branched along B is smooth and integral: the nonempty reduced branch divisor has an odd valuation, so its defining section is not a square in the function

field. Its ramification curves R_i are disjoint (-1) -curves, because $g^*W_i = 2R_i$ and $W_i^2 = -2$. Contract them to a smooth surface $h : Y \rightarrow T$.

Adjunction gives

$$K_S \cdot M = 0, \quad M^2 = -\frac{n}{2}.$$

The standard double-cover decomposition

$$g_*\mathcal{O}_Y = \mathcal{O}_S \oplus \mathcal{O}_S(-M)$$

and Riemann–Roch on the rational surface S give

$$\chi(\mathcal{O}_Y) = 2 - \frac{n}{4}. \quad (5.1)$$

Thus $4 \mid n$. Since every W_i is exceptional for π , one has $M \cdot L = 0$, and hence

$$(K_S + M) \cdot L = K_S \cdot L = -K_X^2 < 0.$$

The nefness of L shows that $K_S + M$ is not effective. Therefore

$$p_g(Y) = h^0(S, K_S) + h^0(S, K_S + M) = 0.$$

Blowing down the R_i does not change p_g or q , so (5.1) yields

$$q(T) = q(Y) = \frac{n}{4} - 1. \quad (5.2)$$

The canonical divisor of Y is not pseudo-effective, because

$$K_Y \cdot g^*L = 2(K_S + M) \cdot L < 0.$$

If K_T were pseudo-effective, then

$$K_Y = h^*K_T + \sum_i R_i$$

would be pseudo-effective as well. Thus K_T is not pseudo-effective. The ruledness criterion [4, Theorem 2.1(I)] shows that T is birationally ruled, with the rational case included.

The double-cover and blow-down formulas give

$$K_Y^2 = 2(K_S + M)^2 = 2K_S^2 - n, \quad K_T^2 = K_Y^2 + n = 2K_S^2.$$

If a smooth surface is birational to a ruled surface over a curve of genus q , then

$$K^2 + \rho = 10 - 8q :$$

the equality holds on a minimal ruled surface, and every blow-up decreases K^2 by one and increases ρ by one. It also holds for rational surfaces, starting from $\mathbb{P}^2$. Writing

$$r = \text{rk}\langle D \rangle = \rho(S) - 1$$

and using $K_S^2 = 10 - \rho(S)$ together with (5.2), we obtain

$$\rho(T) = 2(r - n). \quad (5.3)$$

Every connected component of $D - B$ is a rational tree disjoint from the branch divisor. The restriction of M has degree zero on every irreducible component, hence is trivial by Lemma 2.2. Under such a trivialization, the branch section restricts to a nonzero scalar. That scalar has a square root because the ground field is algebraically closed and the characteristic is not two. The induced étale double cover of the tree therefore splits into two disjoint copies. These copies are disjoint from the ramification curves and survive on T . Since the components of $D - B$ are independent and span a negative-definite lattice of rank $r - n$, the two copies span a negative-definite subspace of $N^1(T)_{\mathbb{Q}}$ of rank

$$2(r - n) = \rho(T),$$

contradicting the Hodge index theorem (if $r = n$, the equality $\rho(T) = 0$ already contradicts projectivity). $\square$

5.1. Divisibility forced by a lattice index. For a nondegenerate integral lattice Γ , let

$$\Gamma^\vee = \{x \in \Gamma \otimes \mathbb{Q} : x \cdot \Gamma \subseteq \mathbb{Z}\}$$

be its dual lattice. Its finite discriminant group is $\Gamma^\vee/\Gamma$, of order $|\det \Gamma|$. For a positive integer I , write $v_2(I)$ for the exponent of 2 in its prime factorization. The Hamming weight $\text{wt}(\epsilon)$ of a binary vector is the number of its nonzero coordinates.

Lemma 5.2 (The full-index obstruction from isolated nodes). *Let S be a smooth projective rational surface and let $\Gamma \subset \text{Pic}(S)$ be a full-rank nondegenerate sublattice. Suppose it has an orthogonal splitting*

$$\Gamma = \Gamma_0 \oplus \bigoplus_{i=1}^t \mathbb{Z}[W_i], \quad W_i^2 = -2,$$

where the W_i are distinct isolated nodes of the original exceptional forest D . Put $I = [\text{Pic}(S) : \Gamma]$. If $t > v_2(I)$, then a nonempty sum of the W_i is twice a divisor in $\text{Pic}(S)$. In particular this inequality is impossible on the rank-one klt del Pezzo resolution of Theorem 5.1. For $\Gamma = \langle D, P \rangle$ with P an exterior (-1) -curve, write $A = A_D$, $p_i = P \cdot D_i$, and $g = p^\top A^{-1} p$. Then

$$I^2 = |\det \Gamma| = \det A (g - 1).$$

Thus the right side must be a positive integer square; every isolated exceptional node disjoint from P may be counted among the t above.

Proof. The Picard lattice of a rational surface is torsion-free and unimodular: this holds on $\mathbb{P}^2$ and every Hirzebruch surface and is preserved by a point blowup, which adds an orthogonal (-1) -summand. Hence $|\det \Gamma| = I^2$. The integral overlattice embeds $G = \text{Pic}(S)/\Gamma$ into $\Gamma^\vee/\Gamma$. Its 2-primary subgroup has order $2^{v_2(I)}$. The projection to $\Gamma_0^\vee/\Gamma_0$ has image of order at most

$$2^{v_2(|\det \Gamma_0|)} = 2^{2v_2(I)-t}.$$

If $t > v_2(I)$, this projection has a nonzero kernel element, supported in the discriminant groups of the W_i alone. It is represented modulo Γ by $\frac{1}{2} \sum_{i \in J} W_i$ for a nonempty subset J . Lifting it to $\text{Pic}(S)$ proves the even-sum assertion. The last determinant formula is the Schur complement of $-A$ in

$$Q_\Gamma = \begin{pmatrix} -A & p \\ p^\top & -1 \end{pmatrix}.$$

The projection identity gives $g > 1$, so this matrix is nondegenerate and has the asserted determinant in absolute value. $\square$

Lemma 5.3 (Riemann–Roch parity and the full Picard index). *Keep the lattice splitting of Lemma 5.2 on a smooth rational surface, and put $s = v_2(I)$. There is a binary linear code $\mathcal{K} \subseteq \mathbb{F}_2^t$ of dimension at least $t - s$ such that for every word $\epsilon \in \mathcal{K}$,*

$$M_\epsilon = \frac{1}{2} \sum_i \epsilon_i W_i \in \text{Pic}(S), \quad K_S \cdot M_\epsilon = 0, \quad M_\epsilon^2 = -\frac{1}{2} \text{wt}(\epsilon).$$

Every word has Hamming weight divisible by four, in every characteristic. Consequently the index data $(I, t) = (36, 4)$ and $(36, 3)$ are impossible already by Riemann–Roch parity. For $(I, t) = (24, 4)$ one has the explicit forced divisibility

$$W_1 + W_2 + W_3 + W_4 = 2M \quad \text{in } \text{Pic}(S), \quad M^2 = -2, \quad K_S \cdot M = 0.$$

The last case is excluded for isolated exceptional nodes in odd positive characteristic by Theorem 5.1.

Proof. The group $G = \text{Pic}(S)/\Gamma$ embeds in $\Gamma^\vee/\Gamma$. Its 2-primary subgroup G_2 has order 2^s . Let $\mathcal{K}$ be the kernel of its projection to $(\Gamma_0^\vee/\Gamma_0)_2$. The latter group has order 2^{2s-t} , because $|\det \Gamma_0| = I^2/2^t$. Thus $|\mathcal{K}| \geq 2^{t-s}$. The kernel lies in the discriminant group $\bigoplus_i \mathbb{Z}/2$ of the node summands, so it is a binary linear code. For each word, lifting to $\text{Pic}(S)$ and subtracting an element of Γ gives exactly the half-sum M_ϵ in that Picard lattice. Adjunction gives $K_S \cdot W_i = 0$, and disjointness gives the displayed square. The standard parity property of

the nodal code recalled above makes its weight divisible by four. For $t = 3$ the only such word is zero. For $t = 4$ the only possible words are zero and $(1, 1, 1, 1)$, so the code has dimension at most one. For $I = 36$ one has $s = 2$: the lower bounds on dimension are respectively one at $t = 3$ and two at $t = 4$, both impossible. For $I = 24$ one has $s = 3$ and $t = 4$, so the nonzero all-one word is forced. Its square and canonical degree are those stated. For isolated exceptional nodes on a rank-one klt del Pezzo resolution in odd positive characteristic, Theorem 5.1 excludes the last case. $\square$

5.2. Lattice bounds for exceptional curve configurations. For the lattice computations below, A_j denotes a chain of j weight-two vertices, and D_4 denotes the weight-two tree with one trivalent vertex and three leaves. The absolute determinants of A_1, A_2, A_3, A_4, D_4 are 2, 3, 4, 5, 4, respectively. For a chain, the determinants satisfy $d_j = 2d_{j-1} - d_{j-2}$, with $d_0 = 1, d_1 = 2$; for D_4 , eliminating its three leaves gives $2^3(2 - 3/2) = 4$.

Lemma 5.4 (An eight-curve weight-two forest has at most four connected components). *Let (S, D, L) be a rational rank-one klt del Pezzo resolution in positive characteristic different from two. Let $F \subset D$ be a union of connected components consisting of eight weight-two curves. Suppose an integral nondegenerate unimodular sublattice $U \subset \text{Pic}(S)$ is orthogonal to F , and $U \oplus \langle F \rangle$ has full rank in $\text{Pic}(S)$. Then F has at most four connected components.*

Proof. If F has at least five components, each has at most four vertices. Its negative definite weight-two trees are therefore A_1, A_2, A_3, A_4 or D_4 , with determinants 2, 3, 4, 5, 4, respectively. The eight possible rank partitions with at least five parts have root determinants

F	$\det F$
$3A_2 + 2A_1$	108
$A_3 + A_2 + 3A_1$	96
$D_4 + 4A_1$	64
$A_4 + 4A_1$	80
$2A_2 + 4A_1$	144
$A_3 + 5A_1$	128
$A_2 + 6A_1$	192
$8A_1$	256

For $\Gamma = U \oplus \langle F \rangle$, unimodularity of $\text{Pic}(S)$ requires $I^2 = |\det \Gamma| = \det F$. Only $D_4 + 4A_1$, $2A_2 + 4A_1$, and $8A_1$ remain. Their pairs (I, t) , where t counts the isolated A_1 components, are $(8, 4), (12, 4), (16, 8)$. In every case $t > v_2(I)$. These are isolated nodes of D and orthogonal summands of Γ , so Lemma 5.2 and Theorem 5.1 exclude all three rows. $\square$

Proposition 5.5 (The zero-adjoint three-curve configuration). *Let (S, D, L) be a minimal rank-one klt del Pezzo resolution in positive characteristic different from two. Suppose $C, B_1, B_2 \subset D$ are pairwise disjoint, $P \not\subset D$ is a (-1) -curve, and*

$$C^2 = -2, \quad B_1^2 = -3, \quad B_2^2 = -\beta \quad (\beta \geq 3), \quad P \cdot C = P \cdot B_1 = P \cdot B_2 = 1, \\ -K_S \equiv C + B_1 + B_2 + 2P.$$

Then $\# \text{Sing}(X) \leq 7$.

Proof. Suppose there are at least eight singular points. Every other exceptional component A satisfies

$$0 \leq K_S \cdot A = -(C + B_1 + B_2 + 2P) \cdot A \leq 0.$$

Hence $A^2 = -2$ and it is disjoint from all four displayed curves. The three displayed exceptional components are isolated, with discrepancies $0, 1/3, (\beta - 2)/\beta$. Since $L \cdot P = (6 - \beta)/(3\beta) > 0$, one has $\beta = 3, 4, 5$. Also $K_S^2 = 3 - \beta$, so $\rho(S) = \beta + 7$. The remaining exceptional curves form an ADE forest F of rank $\beta + 3$ with at least five connected components. The full-rank lattice $\Gamma = \langle C, B_1, B_2, P \rangle \oplus F$ has

$$|\det \Gamma| = (6 - \beta) \det F,$$

since the matrix of its four-curve summand is

$$\begin{pmatrix} -2 & 0 & 0 & 1 \\ 0 & -3 & 0 & 1 \\ 0 & 0 & -\beta & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}, \quad \det = \beta - 6.$$

Unimodularity requires $|\det \Gamma|$ to be a square. The possible rank partitions of F with at least five parts have largest part at most four. Using the determinants 2, 3, 4, 5, 4 of A_1, A_2, A_3, A_4, D_4 leaves exactly the following rows:

β	F	$I = \sqrt{ \det \Gamma }$	$v_2(I)$	$t = \#A_1$
3	$A_2 + 4A_1$	12	2	4
4	$7A_1$	16	4	7
4	$2A_2 + 3A_1$	12	2	3
5	$8A_1$	16	4	8
5	$2A_2 + 4A_1$	12	2	4
5	$D_4 + 4A_1$	8	3	4

Every listed A_1 is an isolated exceptional node and is an orthogonal summand of Γ . Every row has $t > v_2(I)$, contradicting Lemma 5.2 and Theorem 5.1. $\square$

6. ADJOINT DIVISORS AND EXCEPTIONAL CURVE CONFIGURATIONS

An adjoint divisor here is a divisor of the form $K_S + A$, with A specified in each case. The following arguments obtain effective representatives from Riemann–Roch and Serre duality, without a vanishing theorem. A shortest exterior curve means an exterior (-1) -curve of least positive L -degree.

6.1. Configurations with a nef divisor of square one. The three square-one configurations are the following; all displayed curves are smooth rational, P is exterior and all other components belong to D :

type	A	intersections
(U1)	$W + P$	$W^2 = -2, P^2 = -1, W \cdot P = 2$
(U2)	$U + V + P$	$U^2 = V^2 = -2, P^2 = -1, U \cdot V = U \cdot P = V \cdot P = 1$
(U3)	$B + 2P$	$B^2 = -3, P^2 = -1, B \cdot P = 2$

For (U2), require the three displayed intersections to be distinct points. The case where they coincide is also treated by Lemma 6.3.

Lemma 6.1 (The effective adjoint remainder of a square-one configuration). *Let (S, D, L) be a minimal rank-one klt del Pezzo resolution datum, and let A be a square-one divisor centered at $P \notin D$, with all other displayed components of the square-one divisor in D . Put $v = L^2 > 0$ and let $\epsilon = 0$ for (U1), (U2), and $\epsilon = 1$ for (U3). There is an effective integral divisor N_A such that*

$$K_S + A \sim \epsilon P + N_A, \quad L \cdot N_A = L \cdot P - v, \\ \text{Supp}(N_A) \subseteq \text{Null}(A), \quad \text{Supp}(N_A) \cap \text{Supp}(A) = \emptyset.$$

Every exterior A -null curve outside the displayed support occurs in N_A . If these curves are Q_j and their coefficients are n_j , then

$$n_j \in \mathbb{Z}_{>0}, \quad \sum_j n_j (L \cdot Q_j) = L \cdot P - v, \quad 0 < L \cdot Q_j \leq L \cdot P - v < L \cdot P. \quad (6.1)$$

Moreover, $N_A = 0$ if and only if there is no such exterior null curve.

Proof. The surface S is rational by Proposition 2.4. Direct intersection with the three displayed supports gives $A^2 = 1$, $K_S \cdot A = -1$, and respectively the supported degree vectors $(0, 1)$,

$(0, 0, 1)$, and $(1, 0)$. An exterior curve has nonnegative intersection with the effective A , so A is nef. Riemann–Roch therefore gives

$$\chi(\mathcal{O}_S(K_S + A)) = 1 + \frac{1}{2}(K_S + A) \cdot A = 1.$$

Serre duality gives $h^2(K_S + A) = h^0(-A) = 0$: an effective divisor linearly equivalent to $-A$ would have negative intersection with the nef divisor A . Hence choose $Z \geq 0$ with $Z \sim K_S + A$. Since $A \cdot Z = 0$, every component of Z is A -null. Any such component outside $\text{Supp}(A)$ is disjoint from every displayed square-one configuration component.

In (U1), intersecting Z with the only supported null curve W gives $-2 \text{coeff}_W(Z) = 0$. In (U2), the supported coefficients solve

$$\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \text{coeff}_U(Z) \\ \text{coeff}_V(Z) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

so both vanish. In (U3), the only supported null curve is P , and $Z \cdot P = K_S \cdot P = -1$ gives $\text{coeff}_P(Z) = 1$. Thus $Z = \epsilon P + N_A$ has the asserted support. Since $L \cdot K_S = -v$ and $L \cdot A = (1 + \epsilon)L \cdot P$, its L -degree is as stated.

For any exterior null curve Q outside the square-one support, Lemma 2.9 gives $K_S \cdot Q = -1$. It is disjoint from P , so $N_A \cdot Q = -1$. Effectivity forces Q to be a component of N_A , with a positive integral coefficient. All remaining components of N_A lie in D and have L -degree zero, proving (6.1). If there is no exterior component, then N_A is supported on D . For every component C of N_A ,

$$N_A \cdot C = K_S \cdot C = -2 - C^2 \geq 0.$$

Consequently $N_A^2 \geq 0$, contrary to negative definiteness of D unless $N_A = 0$. The converse follows from the occurrence assertion. $\square$

Proposition 6.2 (Singular-point bounds for square-one configurations). *Let (S, D, L) be a minimal rank-one klt del Pezzo resolution datum in positive characteristic different from two. Let $P \not\subset D$ be a smooth rational (-1) -curve with $L \cdot P = \ell_{\text{ext}}$.*

- (i) *If $C \subset D$ is of weight two and $m = C \cdot P \geq 2$, then P is the only exterior null curve of $C + mP$.*
- (ii) *If P centers (U1) or (U2), then $\text{Null}(A) \setminus D = \emptyset$, $K_S + A \sim 0$, $L \sim_{\mathbb{Q}} -K_S$, $L^2 = 1$, and D has eight weight-two components and at most four connected components.*
- (iii) *If $A = B + 2P$ has type (U3), then $\text{Null}(A) \setminus D = \{P\}$, $-K_S \sim B + P$, $L \sim_{\mathbb{Q}} \frac{2}{3}B + P$, and $L^2 = L \cdot P = 1/3$. The curve B is isolated in D ; the other eight exceptional components have weight two and are disjoint from P , and D has at most five connected components.*

In (ii) and (iii), the conclusions also hold without shortestness whenever there is no exterior A -null curve outside its displayed support.

Proof. For (i), $G = C + mP$ is nef and big by its displayed support. An additional exterior null curve Q is a disjoint (-1) -curve by Lemma 2.9. Projecting L to the negative-definite space $G^\perp$ and then to the orthogonal pair P, Q gives

$$(L \cdot Q)^2 \leq \frac{(L \cdot G)^2}{G^2} - L^2 - (L \cdot P)^2 = \frac{2\ell_{\text{ext}}^2}{m^2 - 2} - L^2 < \ell_{\text{ext}}^2,$$

a contradiction. For a square-one configuration, the effective remainder of Lemma 6.1 excludes every additional exterior null curve by shortestness, and gives $N_A = 0$. It gives the same equality directly under the alternative hypothesis.

For (U1) and (U2), $-K_S \sim A$ is nef, and for each D_i adjunction gives $0 \leq A \cdot D_i = 2 + D_i^2 \leq 0$. Thus all exceptional curves have square -2 and all discrepancy coefficients are zero. Since $K_S^2 = 1$, rationality and rank one give $\rho(S) = 9$ and $\# \text{Irr}(D) = 8$. The rank-one sublattice $U = \mathbb{Z}K_S$ has determinant one and is orthogonal to D . Apply Lemma 5.4.

For (U3), $-K_S \sim B + P$. For any other original curve D_i ,

$$0 \leq K_S \cdot D_i = -(B + P) \cdot D_i \leq 0.$$

Hence $D_i^2 = -2$ and it is disjoint from both B and P . The discrepancy vector is $1/3$ on B and zero elsewhere, giving $L \sim_{\mathbb{Q}} \frac{2}{3}B + P$ and $L^2 = L \cdot P = 1/3$. Here $K_S^2 = (B + P)^2 = 0$, so $\rho(S) = 10$ and $D - B$ has eight components. The integral sublattice $U = \langle B, P \rangle$ has matrix

$$\begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$$

and determinant -1 . It is orthogonal to the entire $D - B$. The rank-eight node bound gives at most four connected components there, plus the separate connected component B . $\square$

6.2. Adjacent exceptional curves and multiple intersections.

Lemma 6.3 (The adjacent-contact adjoint). *Work on the minimal resolution datum (S, D, L) . Let $P \not\subset D$ be a smooth rational (-1) -curve and let $C, H \subset D$ be distinct curves with*

$$C^2 = -b, \quad H^2 = -2, \quad b \geq 2, \quad C \cdot H = C \cdot P = H \cdot P = 1.$$

The three intersections need not be distinct points. Then $b \leq 4$. The following divisors A_b are nef and satisfy $A_b^2 = -K_S \cdot A_b = 1$:

b	A_b	$(A_b \cdot C, A_b \cdot H, A_b \cdot P)$	Z_b
2	$C + H + P$	$(0, 0, 1)$	0
3	$C + H + 2P$	$(0, 1, 0)$	P
4	$C + 2H + 3P$	$(1, 0, 0)$	$H + 2P$

There is a unique effective integral divisor N such that

$$K_S + A_b \sim Z_b + N, \quad N \sim K_S + C + H + P, \quad \text{Supp}(N) \cap (C \cup H \cup P) = \emptyset.$$

It is A_b -null and has L -degree $L \cdot P - L^2$. Every exterior A_b -null curve outside the displayed support occurs in N with positive integral coefficient. If $N \neq 0$, its exterior support is nonempty and each exterior component Q satisfies

$$0 < L \cdot Q \leq L \cdot P - L^2 < L \cdot P.$$

In particular $N \neq 0$ if a further exceptional weight-two component $T \neq C, H$ meets P : indeed $N \cdot T = C \cdot T + H \cdot T + P \cdot T > 0$. If $b = 2$ and $N = 0$, then $\#\text{Sing}(X) \leq 7$.

Proof. For $b \geq 3$, the discrepancy equations give $2\lambda_H \geq \lambda_C$ and $b\lambda_C \geq b - 2 + \lambda_H$. Their contribution to the discrepancy sum of P is at least $3(b - 2)/(2b - 1)$, which is at least one for $b \geq 5$. This contradicts $\sum_{D_i} \lambda_i(P \cdot D_i) = 1 - L \cdot P < 1$. Direct intersection gives the table and nefness, since all other curves have nonnegative intersection with the effective A_b . Riemann–Roch and Serre duality give $\chi(K_S + A_b) = 1$ and $h^2(K_S + A_b) = h^0(-A_b) = 0$. Thus choose an effective $Z \sim K_S + A_b$. Every component of Z is A_b -null. Outside the displayed support, it is disjoint from that support. For $b = 2, 3, 4$, the supported coefficients respectively solve

$$\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} z = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -3 & 1 \\ 1 & -1 \end{pmatrix} z = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} -2 & 1 \\ 1 & -1 \end{pmatrix} z = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

These give precisely Z_b and the asserted remainder N . Effectivity and Hodge index imply uniqueness: after removing the common part of two representatives, effective representatives with no common component $N_1 \equiv N_2$ would give $N_1^2 = N_1 \cdot N_2 \geq 0$ in $A_b^\perp$. An exterior null curve is a (-1) -curve by Lemma 2.9; its degree against N is -1 , so it occurs with positive integral coefficient. The L -degree formula then gives the asserted bound. If N had only exceptional components, every component B would satisfy $N \cdot B = K_S \cdot B = -2 - B^2 \geq 0$, contradicting negative definiteness of D unless $N = 0$. The assertion involving T follows by intersection with the remainder class. For $b = 2$, $N = 0$ gives $-K_S \sim C + H + P$ and $K_S^2 = 1$. Rationality and rank one give $\#\text{Irr}(D) = 9 - K_S^2 = 8$. The edge CH is present in the exceptional forest, so its number of connected components is at most seven. $\square$

Corollary 6.4 (The only possible multiple weight-two contact at a shortest curve). *Keep the datum of Lemma 3.3, and let P attain ℓ . If an exceptional weight-two curve C has $m = P \cdot C \geq 2$, then*

$$m = 2, \quad \ell = L^2 = 1, \quad L \sim_{\mathbb{Q}} -K_S, \quad D_i^2 = -2 \quad \text{for every original } D_i.$$

In particular higher weight-two contacts do not occur. The double contact is a type-(U1) square-one configuration and is excluded on a counterexample to Theorem 1.1 by Proposition 6.2.

Proof. The nef class $N = L + \ell K_S$ is orthogonal to both C and P . Their intersection matrix has determinant $2 - m^2 < 0$, so their span contains a positive direction. The orthogonal space of any nonzero nef class on a smooth projective surface is negative semidefinite by Hodge index. Consequently $N \equiv 0$. Its degrees on exceptional components give $K_S \cdot D_i = 0$; hence every D_i has weight two, the discrepancy vector is zero, and $L \sim_{\mathbb{Q}} -K_S$. In particular $\ell = 1$ and $v = L^2 = K_S^2$ is a positive integer. The original projection identity and the Schur bound $(A^{-1})_{CC} \geq 1/2$ now give

$$\frac{m^2}{2} \leq p^\top A^{-1} p = 1 + \frac{1}{v} \leq 2.$$

Thus $m = 2$ and $v = 1$. The displayed curves C, P give the square-one configuration, including a length-two intersection concentrated at one point; Proposition 6.2 applies. $\square$

6.3. Intersections with three disjoint exceptional (-2) -curves.

Proposition 6.5 (An effective adjoint from three (-2) -contacts). *Let $C_0, C_1, C_2 \subset D$ be distinct pairwise disjoint (-2) -curves, and let P be an exterior (-1) -curve with $P \cdot C_i = 1$. Put*

$$F_1 = C_0 + 2P + C_1, \quad F_2 = C_0 + 2P + C_2, \quad A = C_0 + C_1 + C_2 + 3P.$$

Then F_1, F_2 are fiber classes of rational rulings, $F_1 \cdot F_2 = 2$, and A is nef with $A^2 = 3$ and $K_S \cdot A = -3$. There is a unique effective integral divisor E such that

$$E \sim K_S + C_0 + C_1 + C_2 + 2P, \quad \text{Supp}(E) \cap (C_0 \cup C_1 \cup C_2 \cup P) = \emptyset.$$

Its support is null for both F_1 and F_2 , and

$$L \cdot E = 2L \cdot P - L^2. \tag{6.2}$$

Every exterior curve $Q \neq P$ null for both rulings is a (-1) -curve and occurs in E with positive integral coefficient. If $E = 0$, or if there is no such Q , then $n(X) \leq 7$.

Proof. Intersections on the displayed support give $F_i^2 = 0$, $K_S \cdot F_i = -2$, $F_1 \cdot F_2 = 2$, and

$$A^2 = 3, \quad K_S \cdot A = -3, \quad (A \cdot C_0, A \cdot C_1, A \cdot C_2, A \cdot P) = (1, 1, 1, 0).$$

These divisors have nonnegative degree on every other integral curve, so they are nef. The rational-pencil assertion follows from Lemma 3.1. Riemann–Roch gives $\chi(\mathcal{O}_S(K_S + A)) = 1$, while Serre duality gives $h^2(K_S + A) = h^0(-A) = 0$. Choose $Z \geq 0$ with $Z \sim K_S + A$. Since $A \cdot Z = 0$, every component of Z is A -null. The only such component in the displayed support is P ; all others are disjoint from that support. As $Z \cdot P = -1$, the coefficient of P in Z is one. Thus $Z = P + E$ with E as stated.

The support of E is null for $F_1 + F_2$, which is nef and big. Two distinct effective representatives of its class, after subtraction of their common components with the smaller coefficients, would give nonzero effective divisors $E_1 \equiv E_2$ with no common component and $E_1^2 = E_1 \cdot E_2 \geq 0$, contrary to the Hodge index theorem. This proves uniqueness. Intersecting its class with L gives (6.2).

An exterior curve $Q \neq P$ null for both F_i is disjoint from P, C_0, C_1, C_2 and is a (-1) -curve by Lemma 2.9. Hence $E \cdot Q = K_S \cdot Q = -1$, which forces its occurrence in E . If there is no such exterior curve, E is supported on D and every component B of E has $E \cdot B = K_S \cdot B \geq 0$. Negative definiteness forces $E = 0$. In that case $-K_S \sim C_0 + C_1 + C_2 + 2P$ is nef of square two. Adjunction gives $D_i^2 = -2$ for every D_i , so $L = -K_S$ up to rational linear equivalence and $K_S^2 = 2$. Therefore $\# \text{Irr}(D) = 9 - K_S^2 = 7$, proving the bound. $\square$

Proposition 6.6 (A singular-point bound for a single exterior adjoint component). *Work on the resolution datum in positive characteristic different from two. Let $C_0, C_1, C_2 \subset D$ be disjoint weight-two curves meeting the exterior (-1) -curve P once each. Suppose an exterior (-1) -curve R is disjoint from all four curves and*

$$K_S + C_0 + C_1 + C_2 + 2P \sim R. \quad (6.3)$$

Then $\# \text{Sing}(X) \leq 7$.

Proof. Put $p = L \cdot P$, $v = L^2$ and $r = L \cdot R$. The displayed identity gives $K_S^2 = 1$ and $r = 2p - v > 0$. Suppose $\# \text{Sing}(X) \geq 8$. Rationality and rank one give exactly $9 - K_S^2 = 8$ exceptional components, so all eight are isolated in D . Write their squares as $-b_i$, with $b_i \geq 2$. Then

$$v = 1 + \sum_i \frac{(b_i - 2)^2}{b_i}, \quad p = 1 - \sum_i \frac{(b_i - 2)(P \cdot D_i)}{b_i} \leq 1.$$

Since $v < 2p \leq 2$, every b_i is two or three, and the number k of weight-three components belongs to $\{0, 1, 2\}$. Let u be the sum of the contacts of P with those components. Then

$$v = 1 + k/3, \quad p = 1 - u/3, \quad r = 1 - (k + 2u)/3 > 0.$$

For $k = 1, 2$ this forces $u = 0$; for $k = 0$ it is zero by definition. Thus $p = 1$ and P meets no higher-weight component. The rank-one projection identity now gives

$$\frac{1}{2} \sum_{b_i=2} (P \cdot D_i)^2 = 1 + \frac{1}{v} = \begin{cases} 2 & k = 0, \\ 7/4 & k = 1, \\ 8/5 & k = 2. \end{cases}$$

The left side is a half-integer, so $k = 0$ and $p = v = r = 1$. The three weight-two contacts contribute three to the sum of squares, whose value is four. Hence there is exactly one further original component W with $P \cdot W = 1$, and all other contacts of P are zero. Intersecting (6.3) with W gives $R \cdot W = 2$. Thus $W + R$ is a type-(U1) square-one configuration. All discrepancies are zero, so every exterior (-1) -curve has L -degree one and R is shortest. Proposition 6.2 gives at most four singular points, contrary to the assumption of eight. $\square$

Theorem 6.7 (Strict adjoint descent from any three weight-two contacts). *Let P, C_0, C_1, C_2 be as in Proposition 6.6, without assuming its adjoint identity. Then either $\# \text{Sing}(X) \leq 7$, or there is an exterior (-1) -curve Q , disjoint from $P \cup C_0 \cup C_1 \cup C_2$, such that*

$$0 < L \cdot Q \leq L \cdot P - \frac{1}{2}L^2 < L \cdot P. \quad (6.4)$$

It may be chosen from the unique effective divisor $E \sim K_S + C_0 + C_1 + C_2 + 2P$ of Proposition 6.5.

Proof. Put $p = L \cdot P$ and $v = L^2$. The proof of Proposition 6.5 gives $E \geq 0$, disjoint from the four displayed curves, of L -degree $2p - v$. Its support is null for $F_1 + F_2$, where $F_i = C_0 + 2P + C_i$. Hodge index implies uniqueness in its linear class, by the same common-part argument as for a square-one adjoint. Write

$$E = \sum_j a_j Q_j + Z, \quad a_j \in \mathbb{Z}_{>0}, \quad Q_j \not\subset D, \quad Z \geq 0 \text{ on } D, \quad M = \sum_j a_j.$$

The Q_j are (-1) -curves by Lemma 2.9. If $M = 0$, then for every component B of E , $E \cdot B = K_S \cdot B \geq 0$; negative definiteness forces $E = 0$. Then $-K_S \sim C_0 + C_1 + C_2 + 2P$ has square two, and $\# \text{Irr}(D) = 7$, proving the first outcome. If $M = 1$, write $E = R + Z$. Since $E \cdot R = -1$ and $R^2 = -1$, $Z \cdot R = 0$. Hence each component B of Z satisfies $Z \cdot B = E \cdot B = K_S \cdot B \geq 0$, forcing $Z = 0$. Now Proposition 6.6 applies. In all remaining cases $M \geq 2$ and

$$\min_j L \cdot Q_j \leq \frac{\sum_j a_j (L \cdot Q_j)}{M} = \frac{2p - v}{M} \leq p - \frac{v}{2}.$$

Choosing an exterior component of least degree proves the second outcome. $\square$

7. RATIONAL RULINGS WITH SECTIONS AND BISECTIONS

Fix a rational ruling $f : S \rightarrow \mathbb{P}^1$ with fiber class F . A component of D is *vertical* if it lies in a fiber and *horizontal* otherwise. A horizontal component of degree $F \cdot H = 1$ is a section, and one of degree two is a bisection. All fibers in the intersection equations are scheme-theoretic fibers; their component multiplicities are retained.

A *minimal counterexample* is a klt del Pezzo surface of Picard number one with at least eight singular points whose minimal resolution has the smallest Picard number among all such surfaces over the same field. Any birational replacement with at least as many singular points and smaller minimal-resolution Picard number contradicts this choice.

7.1. Connected exceptional divisors and bisection ramification.

Lemma 7.1 (Connected-component count from a rational ruling). *Let*

$$f : S \longrightarrow \mathbb{P}^1$$

be a rational ruling with primitive fiber class F , and write

$$D_h = H_1 + \cdots + H_s, \quad D_v = D - D_h$$

for the horizontal and vertical parts of D . For every reducible fiber

$$\Phi_t = f^{-1}(t)_{\text{red}},$$

write

$$(D_v)_t := \sum_{\substack{C \subset D_v \\ f(C)=t}} C = R_{t,1} + \cdots + R_{t,r_t}$$

as the decomposition into connected reduced divisors; put $r_t = 0$ when the exceptional vertical part is empty. Define

$$a_t = \sum_{i=1}^s \sum_{j=1}^{r_t} H_i \cdot R_{t,j}, \quad q_h = \sum_{1 \leq i < j \leq s} H_i \cdot H_j.$$

Then

$$\#\pi_0(D) = s + \sum_t (r_t - a_t) - q_h. \quad (7.1)$$

Proof. A vertical component of D lies in a reducible fiber: an irreducible fiber has square zero and cannot belong to the negative-definite span of D . In the dual forest of D , contract graph-theoretically each connected subtree carried by one $R_{t,j}$. The resulting graph is a forest with

$$s + \sum_t r_t$$

vertices. Its edges are exactly the horizontal–vertical incidences, of which there are $\sum_t a_t$, and the direct horizontal–horizontal incidences, of which there are q_h . The number of connected components of a forest is its number of vertices minus its number of edges, proving (7.1). $\square$

Lemma 7.2 (Ramification of a smooth bisection). *Let $f : S \rightarrow \mathbb{P}^1$ be a rational ruling over an algebraically closed field of characteristic $p > 0$, and let $B \simeq \mathbb{P}^1$ be a smooth bisection. Put $h = f|_B$ and $\rho_t(B) = \sum_{x \in h^{-1}(t)} (e_x(h) - 1)$, where $e_x(h)$ is the multiplicity in the scheme-theoretic fiber of h . If $p \neq 2$, then h is separable and tame and*

$$\sum_t \rho_t(B) = 2. \quad (7.2)$$

For two bisections in this characteristic one therefore has

$$\sum_t (\rho_t(B_1) + \rho_t(B_2)) = 4. \quad (7.3)$$

If $p = 2$ and h is separable, it has exactly one ramification point, with different exponent two; in this case $\sum_t \rho_t(B) = 1$. If $p = 2$ and h is inseparable, it is Frobenius followed by an

isomorphism, every fiber is twice one point, and the sums of these $\rho_t(B)$ over finite sets of fibers have no uniform upper bound. In every case, if $f^*(t) = \sum_Q m_Q Q$ and $x \in B \cap f^{-1}(t)$, then

$$e_x(h) = \sum_{Q \ni x} m_Q I_x(B, Q).$$

Thus meeting a fiber component of multiplicity at least two forces $e_x(h) = 2$; the total number of such points is at most two when h is separable, but is not so bounded in the inseparable case.

Proof. We use the different form of Riemann–Hurwitz [15, Tag 0C1B]. The finite morphism h has total degree two. Its inseparable degree is a power of p . For $p \neq 2$ it is therefore separable, and every local index is one or two, prime to p . The residue fields are the algebraically closed ground field, so the map is tame. Riemann–Hurwitz makes the sum of the different exponents equal to $-2 - 2(-2) = 2$, and in the tame case these are $e_x - 1$. This proves both equalities. For a separable degree-two map in characteristic two, a ramified local parameter has the form $u = t^2 a(t)$ with $a(0) \neq 0$. The nonzero differential is $t^2 a'(t) dt$, so its order, the different exponent, is at least two. Riemann–Hurwitz still gives total different two. There is consequently exactly one ramification point, with that exponent equal to two and index two. If h is inseparable, its defining rational function belongs to $k(t^2)$ and has degree two. It is therefore $g(t^2)$ with g a fractional linear transformation. This proves the Frobenius description and the fiber assertion; its differential is identically zero, so the separable Riemann–Hurwitz calculation does not apply. Finally restrict the divisor of a local parameter at t from the smooth surface to B . Its order at x is the sum of the component multiplicities times the local intersection numbers, proving the formula. A summand of size at least two uses the entire degree of a fiber of h . $\square$

7.2. Fiber multiplicities and adjoints of bisections.

Lemma 7.3 (Fibers of least possible anticanonical degree). *Let (S, D, L) be the resolution datum of a klt del Pezzo surface of Picard number one in positive characteristic. Suppose exterior (-1) -curves exist and there is a rational ruling with primitive fiber class F satisfying $L \cdot F = 2\ell_{\text{ext}}$. Denote by μ_C the multiplicity of a component C in its scheme-theoretic fiber. In every reducible fiber,*

$$\sum_{R \not\subset D} \mu_R = 2, \quad L \cdot R = \ell_{\text{ext}} \quad (R \not\subset D), \quad C^2 = -2 \quad (C \subset D \text{ vertical}).$$

Thus its exterior multiplicity pattern is (2) or (1, 1). In the first case the exterior fiber component has valency at most two and the exceptional vertical part has at most two connected pieces. In the second case the entire reduced fiber is a chain

$$R - C_1 - \cdots - C_l - Q \quad (l \geq 0),$$

where the C_i are retained (-2) -curves and every multiplicity is one. If s is the number of horizontal components of D , exactly $s - 1$ reducible fibers have pattern (1, 1).

Proof. Every exterior component of a reducible fiber is a smooth rational (-1) -curve, by the fiber intersection equation and adjunction with $K_S = -L - \sum \lambda_C C$. The least-degree inequality and $L \cdot D = 0$ give $\sum \mu_R \leq 2$. On the other hand adjunction against the entire fiber gives

$$-2 = K_S \cdot F = - \sum_{R \not\subset D} \mu_R + \sum_{C \subset D \text{ in the fiber}} \mu_C (-2 - C^2).$$

All terms in the latter sum are nonnegative. Therefore equality holds in both estimates, every retained curve has weight two, and every omitted curve has degree ℓ_{ext} . The reduced fiber is an SNC tree by Lemma 3.2. For a multiplicity-one exterior fiber component the equation $F \cdot R = 0$ says that the sum of its neighbours' multiplicities is one. It is a leaf with a multiplicity-one neighbour. At a retained multiplicity-one (-2) -curve the next neighbour is again unique and of multiplicity one. This traces the whole connected fiber to the other exterior fiber component, proving the chain assertion. A unique exterior fiber component of multiplicity two

has neighbours whose multiplicities sum to two, proving the other bound. Finally, let n_t be the reduced component count and o_t the number of exterior components in the fiber at t . Then

$$\rho(S) = 2 + \sum_t (n_t - 1), \quad \rho(S) - 1 = \# \text{Irr}(D) = s + \sum_t (n_t - o_t).$$

Subtracting gives $\sum_t (o_t - 1) = s - 1$. Each o_t is one or two, so this is the asserted exact number of two-exterior-component fibers. $\square$

Lemma 7.4 (A bisection adjoint is zero or one exterior curve). *Keep the ruling of Lemma 7.3, and let $H \subset D$ be a bisection with $H^2 = -b$. There is an effective integral divisor $N \sim K_S + F + H$, and exactly one of the following holds:*

$$N = 0, \quad \text{or} \quad N = R \text{ for an exterior vertical } (-1)\text{-curve } R \text{ disjoint from } H.$$

In the first case $K_S^2 = 4 - b$, and in the second $K_S^2 = 3 - b$. If $N = 0$, then H is the only horizontal component of D . If $N = R$ and another horizontal component T exists, then the fiber multiplicity of R is one, and every such T satisfies

$$T^2 = -2, \quad T \cdot H = 0, \quad T \cdot R = F \cdot T;$$

its entire fiber intersection is on R .

Proof. Put $A = F + H$. Then $A^2 = 4 - b$, $K_S \cdot A = b - 4$, and $\chi(K_S + A) = 1$. Since $(-A) \cdot F = -2$, Serre duality gives $h^2(K_S + A) = h^0(-A) = 0$. Hence $N \geq 0$ exists. It is vertical, since $N \cdot F = 0$. Its L -degree is $2\ell_{\text{ext}} - L^2 < 2\ell_{\text{ext}}$. An exterior component of an irreducible fiber already has degree $2\ell_{\text{ext}}$; all other exterior vertical components have degree ℓ_{ext} by Lemma 7.3. Thus N has at most one exterior component, with coefficient one. If it has none, it is supported on the weight-two vertical part of D ; for each component C of N one has $N \cdot C = H \cdot C \geq 0$. Negative definiteness forces $N = 0$. Otherwise write $N = R + Z$, with $Z \geq 0$ on D . The equality $N \cdot H = K_S \cdot H + F \cdot H + H^2 = 0$ shows that N is disjoint from H . Hence $N \cdot R = -1$, which gives $Z \cdot R = 0$. For a component C of Z , one has $Z \cdot C = N \cdot C = H \cdot C = 0$. Again negative definiteness gives $Z = 0$. The two identities for K_S^2 follow by squaring $-K_S \sim F + H - N$, using $F \cdot N = H \cdot N = 0$. For another horizontal component T , adjunction gives

$$0 \leq K_S \cdot T = N \cdot T - F \cdot T - H \cdot T.$$

This excludes $N = 0$. For $N = R$, comparison with $F \cdot T \geq \mu_R(R \cdot T)$ forces $\mu_R = 1$ and equality throughout. Thus $K_S \cdot T = H \cdot T = 0$ and all of its fiber intersection lies on R , as claimed. $\square$

7.3. The singular-point bound from sections and bisections.

Theorem 7.5 (The singular-point bound from sections and bisections). *Let (S, D, L) be the resolution datum of a minimal counterexample in positive characteristic different from two. Suppose P is globally shortest among exterior (-1) -curves, and*

$$F = U + 2P + V$$

is a complete rational fiber with $U, V \subset D$ disjoint weight-two curves meeting P once. If $F \cdot C \leq 2$ for all $C \in \text{Irr}(D)$, then $\# \text{Sing}(X) \leq 7$.

Proof. Put $\ell = L \cdot P$. The fiber degree equalities apply since $L \cdot F = 2\ell$. Denote the number of horizontal exceptional components by s , which is positive: otherwise the independent original span of rank $\rho(S) - 1$ and the additional class F would all lie in $F^\perp$. Use the notation r_t, a_t, q_h of Lemma 7.1.

Suppose first that all horizontal components are sections. On the distinguished fiber each meets exactly one of U, V , and Lemma 4.4 gives $s \leq 6$. A one-exterior-component fiber has a multiplicity-two omitted curve, so every section meets its retained part and $a_t = s$, $r_t \leq 2$. The distinguished fiber has $r_0 = 2$, $a_0 = s$. A two-exterior-component fiber has $r_t \leq 1$, so

$r_t - a_t \leq 1$. For $s \geq 2$, all one-exterior-component contributions other than the distinguished fiber are nonpositive. There are $s - 1$ two-exterior-component fibers, giving

$$\#\pi_0(D) = s + \sum_t (r_t - a_t) - q_h \leq s + (s - 1) + (2 - s) = s + 1 \leq 7.$$

For $s = 1$, every reducible fiber is a one-exterior-component fiber and the sole section meets retained support in that fiber. These distinct fibers use distinct edges incident with the section, so there are at most three. Each contributes at most one, giving $\#\pi_0(D) \leq 4$.

Now take a bisection H and its divisor N from Lemma 7.4. If H is the only horizontal component, all reducible fibers have one exterior fiber component of multiplicity two. Outside the at most one fiber supporting N , every retained curve C satisfies $H \cdot C = N \cdot C = 0$. The bisection therefore meets just the exterior fiber component, at one point with scheme-theoretic fiber intersection two. Each such fiber gives one ramification point of the tame degree-two map $H \rightarrow \mathbb{P}^1$. There are at most two by Lemma 7.2. If $N = R$, then on its fiber $H \cdot C = R \cdot C$ for retained C ; thus H meets every retained connected piece of that fiber. Those pieces belong to the connected component containing H . Every other fiber contributes at most two connected components, giving $\#\pi_0(D) \leq 1 + 2 \cdot 2 = 5$. The same bound holds when $N = 0$.

Assume another horizontal component exists. Then $N = R$ and R has fiber multiplicity one. If $H^2 = -2$, then $K_S^2 = 1$ and D has exactly eight components. Unless D has no edges, the desired bound already follows. In the edge-free case the fiber formula gives $2 = F \cdot H = 2P \cdot H$, so P meets the three disjoint weight-two curves U, V, H once each. This contradicts shortestness by Theorem 6.7. Thus we may assume $H^2 \leq -3$. The fiber containing R is the chain $Q - C_1 - \dots - C_l - R$ of Lemma 7.3. Every other horizontal component is of weight two and meets only R . If $l = 0$, intersecting $N = R$ with Q gives $H \cdot Q = 2$. The contact-multiplicity restriction forces $H^2 = -3$, so $H + 2Q$ is a type (U3) square-one configuration at the shortest curve Q . This contradicts Proposition 6.2. For $l \geq 1$, the identities $N = R$ give

$$H \cdot Q = 1, \quad H \cdot C_l = 1, \quad H \cdot C_i = 0 \quad (i < l).$$

The only original contacts of Q are the weight-two curve C_1 and the higher-weight H , once each. Moreover C_1 has valency exactly one in D : its neighbour is C_2 if $l > 1$, and H if $l = 1$. Deleting C_1 makes $Q + (D - C_1)$ an SNC forest, including when the three curves originally met at one point. The unique excess contact of Q is C_1 . The one-component replacement Theorem 4.5 has $s_{C_1} = a = 1$, hence the singular-point count is unchanged and the minimal-resolution Picard number is strictly smaller. Its full contraction theorem therefore gives a Picard descent, contrary to the chosen counterexample. This exhausts the two-contact case. $\square$

8. CURVE CONFIGURATIONS ON A MINIMAL COUNTEREXAMPLE

Assume that $p > 2$ and that a minimal counterexample exists in the sense of section 7. The preceding results now give an exhaustive reduction of the contacts of a shortest exterior (-1) -curve. The conclusion concerns curves on its minimal resolution, not only a numerically admissible graph.

Theorem 8.1 (A single exterior adjoint curve for a minimal counterexample). *Let (S, D, L) be the minimal resolution of a minimal counterexample in positive characteristic different from two. Choose a shortest exterior (-1) -curve P , and put $\ell = L \cdot P$, $v = L^2$. Then there are exceptional curves C, B_1, B_2 and a further original exterior (-1) -curve R satisfying all of the following:*

$$C^2 = -2, \quad (-B_1^2, -B_2^2) = (3, \beta), \quad \beta \in \{3, 4, 5\}, \quad P \cdot C = P \cdot B_1 = P \cdot B_2 = 1.$$

These are all original contacts of P . The curves C, B_1, B_2 are pairwise disjoint; B_1, B_2 lie in different exceptional connected components, and the valency of C in D is zero or one. Moreover

$$R \sim K_S + C + B_1 + B_2 + 2P, \quad R \cap (C \cup B_1 \cup B_2 \cup P) = \emptyset, \quad L \cdot R = 2\ell - v \in [\ell, 2\ell]. \quad (8.1)$$

In particular $0 < v \leq \ell \leq (6 - \beta)/(3\beta)$, and

β	K_S^2	$\# \text{Irr}(D)$	$\#\{\text{edges of } D\}$ at most
3	-1	10	2
4	-2	11	3
5	-3	12	4

For every other original component A one has the contact identity

$$R \cdot A = (-2 - A^2) + (C + B_1 + B_2) \cdot A.$$

At least one edge of D joins $\{C, B_1, B_2\}$ to another component.

Proof. The counterexample has $\rho(S) \geq 9$. A rational surface of this Picard number has a (-1) -curve, which is exterior because D is the minimal exceptional divisor. The shortest degree exists by discreteness. By Corollary 6.4 and Proposition 6.2, every weight-two contact of P is simple. The excess-contact lemma and the same square-one bound make every higher-weight contact simple as well: any multiple one would have weight three and contact two, giving type (U3). There are at most two higher-weight contacts, since each contributes at least $1/3$ to the discrepancy sum, which is less than one. Two in the same original component are excluded by Lemma 4.3. An excess contact exists, so there is a weight-two contact. Two adjacent weight-two contacts are excluded by Lemma 6.3: its nonzero remainder produces a shorter curve, while its zero-remainder $b = 2$ case gives the count. Three disjoint ones are excluded by Theorem 6.7.

Suppose there are two weight-two contacts U, V . Their divisor $F = U + 2P + V$ is nef, square zero and of canonical degree -2 . Its primitivity follows from Riemann–Roch parity, so Lemma 3.1 gives its complete rational pencil. A contacted higher-weight exceptional curve B cannot meet U : applying Lemma 6.3 to B, U, P , the other end V has positive degree against its remainder and forces a shorter curve. Likewise B cannot meet V . Hence $F \cdot B = 2$. Every other original component misses P , and its F -degree is at most two by the SNC property of D . Thus every horizontal exceptional curve has degree at most two, and Theorem 7.5 excludes this case.

There is therefore exactly one weight-two contact C . If j is the number of higher-weight contacts, deleting C leaves $P + (D - C)$ an SNC forest: the remaining contacts are simple and lie in distinct exceptional components. The one-component replacement changes the number of singular points by $s_C - 1$ for $j = 0$ and $s_C - j$ for $j > 0$. Any nonnegative change contradicts Picard minimality. For $j = 0$ the remaining possibility is one simple isolated- A_1 contact, whose projection quadratic form $1/2$ is impossible. For $j = 1$ the remaining possibility is the isolated- A_1 mixed exchange, excluded by Theorem 4.6. Thus $j = 2$ and $s_C \leq 1$; denote the other contacts by B_1, B_2 . The curve C cannot be adjacent to B_i : the adjacent-contact remainder would have positive intersection with the other B_j , because $K_S \cdot B_j \geq 1$ and $P \cdot B_j = 1$. That remainder is nonzero and would again give a shorter curve. The strict discrepancy-sum inequality and $\lambda_{B_i} \geq 1 - 2/(-B_i^2)$ give $1/(-B_1^2) + 1/(-B_2^2) > 1/2$. After relabelling, this gives exactly $(3, 3), (3, 4), (3, 5)$.

Put $T = C + B_1 + B_2 + 2P$. Then $T^2 = 3 - \beta$, $K_S \cdot T = \beta - 3$, so $\chi(K_S + T) = 1$ and $h^2(K_S + T) = h^0(-T) = 0$. Choose $N \geq 0$ with $N \sim K_S + T$. It satisfies

$$N \cdot P = N \cdot C = N \cdot B_1 = N \cdot B_2 = 0, \quad L \cdot N = 2\ell - v < 2\ell.$$

By Lemma 3.3, its total exterior coefficient is zero or one. If zero, N is supported on D and $N^2 = K_S \cdot N \geq 0$, since $N \cdot T = 0$. Negative definiteness gives $N = 0$, contrary to Proposition 5.5. Thus $N = R + Z$ for an exterior prime R and $Z \geq 0$ on D . If $R = P$, the vanishing of N against each of C, B_1, B_2 forces all three to occur in Z ; then $Z \cdot P \geq 3$, contrary to $N \cdot P = -1 + Z \cdot P = 0$. Hence $R \neq P$. The equality $N \cdot P = 0$ now gives $R \cdot P = Z \cdot P = 0$, so none of C, B_1, B_2 occurs in Z . Vanishing against those three curves forces both R and Z to be disjoint from them. Therefore N is disjoint from $\text{Supp}(T)$. Now $N \cdot R = K_S \cdot R < 0$ implies $R^2 < 0$; adjunction makes R a (-1) -curve. It follows that $Z \cdot R = 0$. For every component A

of Z , one then has $Z \cdot A = N \cdot A = K_S \cdot A \geq 0$; hence $Z = 0$. This proves (8.1). Its degree lower bound is shortestness, and its upper bound follows from the elementary discrepancies of B_1, B_2 . Squaring $K_S \sim R - T$ gives $K_S^2 = 2 - \beta$. Rationality gives $\# \text{Irr}(D) = \beta + 7$, and the forest count with at least eight components gives at most $\beta - 1$ edges. Intersecting the adjoint identity with any other original A , using that P has no further original contacts, proves the contact formula. If there were no edge from the three original core curves to the rest of D , this formula would make every contact of R bounded. That contradicts Lemma 4.2, proving the last assertion. $\square$

9. EXCEPTIONAL DUAL GRAPHS AND THE SEVEN-POINT BOUND

The reduction in Theorem 8.1 bounds the total number of edges. We first classify the required rational intersection matrices, and then apply the result to the exceptional curves of a minimal counterexample.

9.1. Inverse matrix entries for small weighted trees.

Lemma 9.1 (Inverse matrix entries for trees with one higher-weight vertex). *Let a positive definite weighted tree have one vertex of weight $b \geq 3$ (the root) and all other vertices of weight two. Assume that its valencies are at most three and it has at most three edges. If α is the root diagonal entry of its inverse positive matrix, the complete list is*

edges	rooted tree	α
0	$[b]$	$1/b$
1	$[b, 2]$	$2/(2b - 1)$
2	$[b, 2, 2]$	$3/(3b - 2)$
2	$[2, b, 2]$	$1/(b - 1)$
3	$[b, 2, 2, 2]$	$4/(4b - 3)$
3	$[2, b, 2, 2]$	$6/(6b - 7)$
3	three-leaf star, root at the centre	$2/(2b - 3)$
3	three-leaf star, root at a leaf	$1/(b - 1)$

For the canonical-degree vector $q = (b - 2)e_{\text{root}}$, the root coefficient and correction are $(b - 2)\alpha$ and $(b - 2)^2\alpha$. For two disjoint weight-three rooted blocks let τ be the sum of their root diagonal entries. With at most one or two total edges its possible values are respectively

$$\mathcal{S}_1 = \{2/3, 11/15\}, \quad \mathcal{S}_2 = \mathcal{S}_1 \cup \{16/21, 5/6, 4/5\}.$$

With at most three total edges, the values satisfying $13/15 < \tau \leq 14/15$ are exactly $29/33$ and $9/10$. An all-weight-two component with a distinguished leaf has inverse entry $1/2, 2/3, 3/4$ for zero, one, two edges, respectively; with three edges its possible entries are $4/5$ and 1 .

Proof. With at most three edges the only unrooted trees are a vertex, a path, and the three-leaf star. Their root positions give precisely the eight rows. Eliminating weight-two leaves uses $a_i \leftarrow a_i - 1/a_j$. A weight-two chain of length r attached to the root contributes $r/(r + 1)$ to its Schur complement. Thus the path rows and the centred star follow immediately; for the leaf-rooted star the remaining weight-two three-vertex path has centre inverse entry one. The two fiber formulas follow by pairing $(b - 2)e_{\text{root}}$ with the inverse. At $b = 3$, the possible root inverse entries for exactly 0, 1, 2, 3 edges are

$$\{1/3\}, \quad \{2/5\}, \quad \{3/7, 1/2\}, \quad \{4/9, 6/11, 2/3, 1/2\}.$$

Adding two such entries with the stated total edge bound gives $\mathcal{S}_1, \mathcal{S}_2$. For three edges the only new sums are $7/9, 29/33, 1, 29/35, 9/10$, giving the indicated interval intersection. Applying the same elimination to the weight-two leaf gives the last assertion, including the leaf of the weight-two three-leaf star. $\square$

9.2. The ten candidate exceptional dual graphs.

Lemma 9.2 (Classification of the candidate weighted forests). *Let $\beta \in \{3, 4, 5\}$ and $n = \beta + 7$. Consider an n -vertex weighted forest with three distinguished vertices C, B_1, B_2 of weights $2, 3, \beta$, respectively. Assume:*

- (a) *every other weight is two or three, and at most two other vertices have weight three;*
- (b) *the distinguished vertices are pairwise nonadjacent, B_1, B_2 lie in different connected components, $\deg C \leq 1$, and every vertex has valency at most three;*
- (c) *there are at most $\beta - 1$ edges, and at least one edge joins a distinguished vertex to another vertex.*

Let A have these weights on the diagonal and entries -1 at the edges. Let $q = \text{diag}(A) - 2\mathbf{1}$ and let p have entries one on C, B_1, B_2 and zero elsewhere. Suppose A is positive definite and, with

$$\lambda = A^{-1}q, \quad \ell = 1 - p^\top \lambda, \quad v = 2 - \beta + q^\top \lambda, \quad g = p^\top A^{-1}p,$$

one has

$$0 \leq \lambda_i < 1, \quad 0 < v \leq \ell, \quad v(g - 1) = \ell^2.$$

Then the forest is exactly one of the ten rows below. In the first table $[\dots]$ denotes a chain, a bare numeral denotes an undistinguished vertex of that weight, and A_j denotes an undistinguished weight-two chain of rank j .

Family	β	Fixed connected components	Remaining forest Z
A	3	$[C, 3] + [B_1] + [B_2]$	$6A_1$ or $A_2 + 4A_1$
B	3	$[C, 2, 3] + [B_1] + [B_2]$	$5A_1$
C	3	$[C] + [B_1, 2] + [B_2, 2] + [3]$	$4A_1$
D	3	$[C, 2] + [B_1] + [B_2] + 2[3]$	$4A_1$ or $A_2 + 2A_1$
E	4	$[C] + [B_1, 2] + [B_2] + 2[3]$	$5A_1, A_2 + 3A_1,$ $A_3 + 2A_1, \text{ or } 2A_2 + A_1$

Number multiple completions in their displayed order: $A1, A2$; $D1, D2$; and $E1$ through $E4$. The family data are given below. Here $\Delta = \det A(g - 1)$, so the last column is multiplied by the root determinant of the stated Z .

family	ℓ	v	g	$\Delta / \det Z$
A	2/15	1/15	19/15	12
B	4/21	2/21	29/21	24
C	1/5	2/15	13/10	45
D	1/3	1/3	4/3	81
E	1/10	1/15	23/20	54

The absolute full-lattice determinants for the ten rows are

row	A1	A2	B	C	D1	D2	E1	E2	E3	E4
Δ	768	576	768	720	1296	972	1728	1296	864	972.

In particular no row has $\beta = 5$.

Proof. Write $T_1, \dots, T_k$ for the extra weight-three vertices, $0 \leq k \leq 2$. Call a vertex of weight at least three higher-weight. In addition to $\ell > 0$, use the following explicit form of $v \leq \ell$:

$$v - \ell = 1 - \beta + \lambda_C + 2\lambda_{B_1} + (\beta - 1)\lambda_{B_2} + \sum_i \lambda_{T_i} \leq 0. \quad (9.1)$$

If $k = 0$, then $v = 2 - \beta + \lambda_{B_1} + (\beta - 2)\lambda_{B_2} < 0$, since $\lambda_{B_1} + \lambda_{B_2} < 1$ and $\beta - 2 \geq 1$. Thus $k = 1$ or $k = 2$.

Each connected component has at most one higher-weight vertex. For any path, restriction of the full discrepancy equations gives its own equations with a nonnegative additional vector. Stieltjes positivity therefore bounds its coefficients from below by those of the path alone. If two weight-three vertices are joined through weight-two vertices, both endpoint coefficients are at least $1/2$: the constant vector $1/2$ is exactly the path solution. If B_1 and some T_i belong to

one component, take the path to the nearest other higher-weight vertex. It cannot be B_2 . For $\beta = 3$, the two $1/2$ coefficients and $\lambda_{B_2} \geq 1/3$ make the right side of (9.1) at least $1/6$. For $\beta \geq 4$ they give $\lambda_{B_1} + \lambda_{B_2} \geq 1$, contradicting $\ell > 0$. If B_2 is joined to a nearest T_i by a path of $d \geq 1$ edges, let $D_d = 2\beta d + \beta - 2d + 1$. On that path the coefficients at the intervening weight-two vertices form an arithmetic progression. If the endpoint coefficients are u, w , their endpoint equations are $(\beta - 1)u + (u - w)/d = \beta - 2$ and $2w + (w - u)/d = 1$. Solving gives the lower bounds

$$\lambda_{B_2} \geq \frac{(\beta - 2)(2d + 1) + 1}{D_d}, \quad \lambda_{T_i} \geq \frac{(\beta - 1)(d + 1)}{D_d}.$$

Together with $\lambda_{B_1} \geq 1/3$, these make $v - \ell$ at least

$$\frac{\beta d - \beta - d + 5}{3D_d} > 0.$$

Thus neither core higher-weight vertex shares a component with another higher-weight vertex. Finally, if the two extra higher-weight vertices share a component, their coefficients are at least $1/2$ each; the isolated core lower bounds make $v - \ell \geq (6 - \beta)/(3\beta) > 0$.

The component of C . Suppose first it contains a core higher-weight vertex. The path has $d \geq 2$ edges. If that vertex is B_1 , the endpoint sum is at least $(d + 2)/(2d + 3) > 1/2$; for $\beta \geq 4$ the other core coefficient is at least $1/2$, contradicting $\ell > 0$. For B_2 the endpoint sum is at least $(\beta - 2)(d + 2)/((\beta - 1)d + \beta) > 2/3$ when $\beta \geq 4$, again contradicting the remaining coefficient $\lambda_{B_1} \geq 1/3$. For $\beta = 3$ the edge bound forces the entire path $[C, 2, B_i]$ and leaves every other vertex isolated. Its values are

$$\ell = 2/21, \quad g = 37/21, \quad v = (7k - 5)/21.$$

For $k = 2$ one has $v > \ell$; for $k = 1$ the quantity $v(g - 1) - \ell^2 = 4/63$ is nonzero. This case is excluded.

Next suppose C shares a component with T_i , at distance d . The path gives $\lambda_C \geq 1/(2d + 3)$ and $\lambda_{T_i} \geq (d + 1)/(2d + 3)$. For $\beta = 5$, $d \leq 4$ makes $\lambda_C \geq 1/11 > 1/15$, whereas the isolated core lower bounds require $\lambda_C < 1/15$. For $\beta = 4$, $k = 2$, these path bounds and the other extra coefficient at least $1/3$ make $v - \ell \geq (d + 2)/(2d + 3) - 1/2 > 0$. For $\beta = 4$, $k = 1$, the fact that C is a leaf gives $\lambda_{T_i} \leq (d + 1)\lambda_C \leq 4\lambda_C$: along the weight-two path its coefficient increments are nonincreasing, since every side branch contributes a nonnegative term. Then $\ell > 0$ implies

$$v = -2 + \lambda_{B_1} + 2\lambda_{B_2} + \lambda_{T_i} < 2 - 3\lambda_{B_1} - 2\lambda_{B_2} \leq 0.$$

For $\beta = 3$, $k = 2$, the same path lower bounds give $v - \ell \geq (d + 2)/(2d + 3) - 1/3 > 0$. It remains $\beta = 3$, $k = 1$ with at most two edges. The only possible component of C is $[C, 3]$, $[C, 2, 3]$, or $[C, 3, 2]$. The third has $(\ell, v) = (1/12, 1/6)$ and is excluded. For $[C, 3]$ the core vertices are isolated: attaching the possible second edge to either B_i would increase v and decrease ℓ by $1/15$, violating $v \leq \ell$. The remaining edge is therefore an unrelated weight-two edge, if present. This gives precisely family A. The component $[C, 2, 3]$ uses both edges and gives family B. Direct inversion gives their displayed numerical data.

For all remaining cases the three distinguished vertices lie in distinct components, and C 's component has weight two. Put

$$x = (A^{-1})_{B_1 B_1} - 1/3, \quad y = (A^{-1})_{B_2 B_2} - 1/\beta, \quad z = (A^{-1})_{CC}, \quad \tau = \sum_i (A^{-1})_{T_i T_i}.$$

These quantities come from the separate inverse-matrix blocks. In particular $x, y \geq 0$, $z \geq 1/2$, and $\tau \geq k/3$. The three scalar identities become

$$\ell = 2/\beta - 1/3 - x - (\beta - 2)y, \tag{9.2}$$

$$v = -5/3 + 4/\beta + x + (\beta - 2)^2 y + \tau, \tag{9.3}$$

$$g = z + 1/3 + 1/\beta + x + y. \tag{9.4}$$

The inequality $v \leq \ell$ is

$$\tau + 2x + (\beta - 2)(\beta - 1)y \leq 4/3 - 2/\beta. \quad (9.5)$$

$\beta = 3$. For $k = 2$, (9.5) forces $x = y = 0$, $\tau = 2/3$. All four higher-weight vertices are isolated. Then $v = \ell = 1/3$, and the projection identity forces $z = 2/3$. Thus C 's component is exactly $[C, 2]$; the possible remaining edge is an unrelated weight-two edge. These are the two rows of family D. For $k = 1$, put $s = x + y$ and $u = \tau - 1/3$. The projection identity becomes

$$s(z + 1/3) + u(z - 1/3 + s) = 1/9. \quad (9.6)$$

If C is isolated, $z = 1/2$. For $u = 0$, this forces $s = 2/15$. With at most two edges at the components containing B_1 and B_2 the possible positive sums s are $1/15, 2/21, 1/6, 2/15$; only one weight-two leaf at each B_i gives the required value. This is family C. For $u > 0$, the component containing the extra weight-three vertex has an edge; the core-boundary condition requires another edge at a B_i . Hence $s = u = 1/15$, and the left side of (9.6) is $16/225 \neq 1/9$. If C is not isolated, positivity of $v = s + u$ requires an edge at a component containing a higher-weight vertex as well. The two-edge bound therefore forces $z = 2/3$ and $(s, u) = (1/15, 0)$ or $(0, 1/15)$. The left side is then $1/15$ or $1/45$, neither of which is $1/9$. This finishes $\beta = 3$.

$\beta = 4$. For $k = 1$, the discrepancy-sum inequality $x + 2y < 1/6$ makes each distinguished higher-weight component a path with its higher-weight vertex at an end. Indeed every other root placement in the at-most-three-edge table has root inverse entry at least $1/2$ for B_1 , or at least $1/3$ for B_2 , contradicting that inequality. Let their numbers of edges be i, j , so

$$x = \frac{i}{3(2i + 3)}, \quad y = \frac{j}{4(3j + 4)}.$$

For $d = i + j \leq 3$, maximizing over its $d + 1$ possibilities subject to $x + 2y < 1/6$ gives the first row below. The second bounds the remaining inverse entry at the extra weight-three vertex using at most $3 - d$ edges:

d	0	1	2	3
$\max(x + 4y)$	0	$1/7$	$22/105$	$3/13$
$\max \tau$	$2/3$	$1/2$	$2/5$	$1/3$

Each column sum is at most $2/3$, so $v = -2/3 + x + 4y + \tau \leq 0$. This excludes $k = 1$. For $k = 2$, (9.5) gives $2x + 6y \leq 1/6$. An edge at B_2 gives $y \geq 1/28$, which is impossible. Two edges in B_1 's component give $x \geq 2/21$, also impossible. Thus B_2 is isolated and B_1 is isolated or has one weight-two leaf. In the latter case $x = 1/15$, so $\tau \leq 7/10$. An edge at a component containing an extra weight-three vertex would give $\tau \geq 11/15$; hence both extras are isolated. Now $(\ell, v) = (1/10, 1/15)$ and the projection identity forces $z = 1/2$. The five remaining weight-two vertices have at most two edges, giving exactly the four completions of family E. If both core higher-weight vertices are isolated, the required core boundary edge is in the weight-two component of C . With one edge there, $z = 2/3$ and the projection identity forces $\tau = 7/9$, which is not in $\mathcal{S}_2$. With two edges, $z = 3/4$ and the projection identity forces $\tau = 3/4 \notin \mathcal{S}_1$. With three edges no edge remains at a component containing an extra weight-three vertex, and $v = 0$. There are no further rows.

$\beta = 5$. The positive length bound is $x + 3y < 1/15$. Even one edge at B_1 gives $x \geq 1/15$; one edge at B_2 gives $3y \geq 1/15$. Thus both are isolated and the component of C has an edge. For $k = 1$, the component containing the extra weight-three vertex has at most three edges and $\tau \leq 2/3$, so $v = \tau - 13/15 < 0$. For $k = 2$, if the component of C has at least two edges, the two components containing T_1, T_2 have at most two total edges, giving $\tau \leq 5/6 < 13/15$ and again $v < 0$. The sole remaining case has $[C, 2]$, hence $z = 2/3$, $\ell = 1/15$ and $g = 6/5$. The projection identity requires $v = 1/45$ and $\tau = 8/9$. But $0 < v \leq \ell$ requires $13/15 < \tau \leq 14/15$, and the only such values with at most three remaining edges are $29/33$ and $9/10$, neither of which is $8/9$. Thus $\beta = 5$ is impossible.

This determines the fixed components of all five families. For family A there are six remaining weight-two vertices and at most one remaining edge, giving $6A_1$ or $A_2 + 4A_1$. Family

B leaves five isolated vertices, and family C leaves four isolated vertices. Family D leaves four weight-two vertices with at most one edge, giving $4A_1$ or $A_2 + 2A_1$. Family E leaves five weight-two vertices with at most two edges; their edge components are empty, a single edge, a three-vertex path, or two disjoint edges. This gives precisely the four displayed completions. No other connected shape can occur with these edge allowances. Inverting their fixed blocks gives the stated ℓ, v, g and $\Delta/\det Z$; weight-two components of Z contribute only their usual root determinants. This proves the full ten-row classification. $\square$

9.3. Determinant, parity, and double-cover obstructions.

Theorem 9.3 (The single exterior adjoint configuration is impossible). *The configuration of Theorem 8.1 cannot occur on a minimal counterexample in positive characteristic different from two. All three values $\beta = 3, 4, 5$ are excluded.*

Proof. Retain all the exceptional curves and put $\ell = L \cdot P$ and $v = L^2$. For another original component A of weight $b = -A^2$, let $u = (C + B_1 + B_2) \cdot A \geq 0$. The adjoint identity gives

$$R \cdot A = b - 2 + u.$$

The elementary discrepancy bound and the strict discrepancy-sum inequality of the actual (-1) -curve R therefore imply

$$1 > \sum_{D_i} \lambda_i(R \cdot D_i) \geq \frac{(b-2)^2}{b}.$$

For $b \geq 4$ the last expression is at least one, a contradiction. Thus every additional original weight is two or three. Every additional weight-three vertex contributes at least $1/3$ to that same sum, so there are at most two of them. The valency-three bound is Lemma 4.4. All other graph requirements of Lemma 9.2 are conclusions of Theorem 8.1.

For the full negative intersection matrix A_D , its canonical-degree vector q , and the contact vector p of P , the original identities are

$$\lambda = A_D^{-1}q, \quad \ell = 1 - p^\top \lambda, \quad v = K_S^2 + q^\top \lambda = 2 - \beta + q^\top \lambda, \quad v(p^\top A_D^{-1}p - 1) = \ell^2.$$

Here $0 < v \leq \ell$ is already proved by shortestness of P and the degree $L \cdot R = 2\ell - v$. Thus the complete finite matrix table applies, without assuming the additional curves have weight two or assuming the table realizable. The sublattice $\Gamma = \langle D, P \rangle \subset \text{Pic}(S)$ has full rank and determinant Δ as in that table. Unimodularity rejects every row whose Δ is not a square. Only A2, D1, and E2 remain:

row	Δ	$I = \sqrt{\Delta}$	$v_2(I)$	t
A2	576	24	3	4
D1	1296	36	2	4
E2	1296	36	2	3

The t indicated vertices are isolated weight-two components of D disjoint from P , hence orthogonal node summands of Γ . By Lemma 5.3, D1 and E2 violate Riemann–Roch parity on any smooth rational surface. In A2, writing its four isolated nodes as $W_1, \dots, W_4$, the explicit forced class is $M = (W_1 + W_2 + W_3 + W_4)/2 \in \text{Pic}(S)$ with $M^2 = -2$ and $K_S \cdot M = 0$. It contradicts Theorem 5.1 in odd positive characteristic. This excludes every row. $\square$

9.4. Proof of the seven-point bound.

Proof of Theorem 1.1. Fix an algebraically closed field k of characteristic $p > 2$ and suppose that there is a counterexample. Choose one whose minimal resolution S has the least Picard number. By Proposition 2.4, S is rational and its exceptional divisor D is a simple-normal-crossing rational forest with $\# \text{Irr}(D) = \rho(S) - 1 \geq 8$. In particular $\rho(S) > 2$. The nef-threshold lemma, Lemma 3.3, gives a shortest exterior (-1) -curve and controls the degree of every exterior integral curve.

All replacements used in Theorem 8.1 are defined over k . Their projectivity and klt anticanonical contractions follow from Theorems 2.6 and 2.7; whenever they do not decrease the number of singular points, they contradict the minimality of $\rho(S)$. The adjoint arguments are intersection and Riemann–Roch arguments on this rational surface. The discrepancy coefficients and matrix inversions are computed over $\mathbb{Q}$; their denominators impose no further condition on the characteristic of k . The bisection argument and the exceptional-node double cover are valid because $p \neq 2$. Therefore Theorem 8.1 gives the stated single exterior adjoint configuration. Its exclusion in Theorem 9.3 is a contradiction. Thus $n(X) \leq 7$. $\square$

10. FROBENIUS CONSTRUCTIONS AND SHARPNESS

The characteristic-two family is due to Keel–McKernan [7, Section 9], and the characteristic-three seven-point surface is Bernasconi’s [2, Section 3.1 and Proposition 3.2]; see also [9, Example 7.4]. The Frobenius graph and the successive contact blowups below recover Bernasconi’s construction exactly at $(p, n) = (3, 3)$. Bernasconi also observes the higher-characteristic change of canonical sign [2, Remark 3.3].

Bernasconi contracts the configuration by a surface minimal model program; here an explicit nef divisor and its exact null locus give the contraction directly by Keel’s theorem. The calculation treats arbitrary p, n , and supplies the Picard classes needed for the subsequent analysis of degree-two maps and shortest-curve replacements. It does not use the upper-bound theorem.

Proposition 10.1 (Canonical divisors in the Frobenius family). *Let k be algebraically closed of characteristic $p > 0$, and let $B_0 \subset \mathbb{P}_x^1 \times \mathbb{P}_y^1$ be the graph $y = x^p$. Write a, b for the fiber classes $x = \text{constant}$ and $y = \text{constant}$. Choose $n \geq 3$ distinct points of the graph. At each one blow up p times along its contact with the corresponding b -fiber, obtaining $S_{p,n}$ with total-transform classes E_{ij} . The strict transforms and exceptional curves have the following classes:*

$$B = pa + b - \sum_{i,j} E_{ij}, \quad F_i = b - \sum_{j=1}^p E_{ij}, \quad C_{ij} = E_{ij} - E_{i,j+1} \quad (1 \leq j < p), \quad P_i = E_{ip}.$$

The reduced divisor $D = B + \sum_i (F_i + \sum_{j < p} C_{ij})$ consists of one isolated curve of square $-p(n-2)$, n isolated curves of square $-p$, and n disjoint weight-two chains A_{p-1} . It contracts projectively to a normal klt surface $X_{p,n}$ with $\rho(X_{p,n}) = 1$ and $\#\text{Sing}(X_{p,n}) = 2n + 1$. Denote this contraction by $\pi : S_{p,n} \rightarrow X_{p,n}$. Put

$$M = B + (n-2)b, \quad d_{p,n} = 2 - (p-2)(n-2), \quad t_{p,n} = \frac{d_{p,n}}{p(n-2)}.$$

Then M is nef and big with exact null locus D , and

$$M^2 = p(n-2), \quad L := \pi^*(-K_{X_{p,n}}) \sim_{\mathbb{Q}} t_{p,n}M, \quad L^2 = \frac{d_{p,n}^2}{p(n-2)}, \quad L \cdot P_i = t_{p,n}.$$

The coefficients in the canonical pullback on B, F_i, C_{ij} are, respectively,

$$1 - \frac{2}{p(n-2)}, \quad 1 - \frac{2}{p}, \quad 0.$$

In particular $-K_{X_{p,n}}$ is ample exactly for $p = 2$ (all $n \geq 3$) or for $(p, n) = (3, 3)$; $K_{X_{3,4}} \sim_{\mathbb{Q}} 0$; and $K_{X_{p,n}}$ is ample in all other cases with prime p and $n \geq 3$. For $p = 3$ we also write $S_n, X_n, D_n, M_n, t_n, L_n$ and set $U_i = C_{i1}, V_i = C_{i2}$, recovering the seven-point notation below.

Proof. At a finite selected point (c, c^p) put $u = x - c, v = y - c^p$. The graph is $v = u^p$. In the successive charts $v = u^j v_j$ its strict transform is $v_j = u^{p-j}$ and the strict fiber is $v_j = 0$. For $j < p$ they meet at the next blowup center; at $j = p$ the graph, fiber, and previous exceptional curve meet P_i at $v_p = 1, 0, \infty$, respectively. Earlier exceptional curves form the displayed chain.

Reciprocal coordinates $u = 1/x$, $v = 1/y$ give the same equations at infinity. Thus the strict curves are smooth and the only local incidences are

$$B \cdot P_i = F_i \cdot P_i = C_{i,p-1} \cdot P_i = 1, \quad C_{ij} \cdot C_{i,j+1} = 1.$$

In particular all blocks of D are disjoint negative definite rational trees. There are $np + 1 = \rho(S_{p,n}) - 1$ independent exceptional components. The total-transform basis gives $K_{S_{p,n}} = -2a - 2b + \sum E_{ij}$. Let $\lambda_B = 1 - 2/[p(n-2)]$ and $\lambda_F = 1 - 2/p$. Solving the canonical intersection equations on these disjoint blocks gives these coefficients on B, F_i , and zero on all C_{ij} . All belong to $[0, 1)$. Define the rational divisor

$$\Lambda = - \left(K_{S_{p,n}} + \lambda_B B + \lambda_F \sum_i F_i \right).$$

Substitution of the displayed classes gives $\Lambda = t_{p,n}M$ in $\text{Pic}(S_{p,n})_{\mathbb{Q}}$.

Direct intersection gives $M \cdot B = 0$ and $M^2 = p(n-2) > 0$. If an irreducible curve $R \neq B$ is not vertical for the b -ruling, then $M \cdot R = B \cdot R + (n-2)b \cdot R > 0$. A nonspecial vertical fiber has degree p . A special scheme-theoretic fiber is

$$F_i + \sum_{j=1}^{p-1} j C_{ij} + p P_i.$$

Here $M \cdot P_i = 1$ and its other components have degree zero. This proves nefness and the exact null locus. The restriction to each original rational tree is trivial, so Theorem 2.3 and Lemma 2.2 give a projective Stein contraction $\pi : S_{p,n} \rightarrow X_{p,n}$ with this exact exceptional locus. It is birational since M is big and the Stein map has equal function fields. Its target is normal; the independent exceptional rank and an ample target class give Picard number one. Write $M \sim_{\mathbb{Q}} \pi^* A$ with A ample. Pushforward of the canonical identity gives $K_{X_{p,n}} \sim_{\mathbb{Q}} -t_{p,n}A$, hence $K_{X_{p,n}}$ is $\mathbb{Q}$ -Cartier. The divisor $K_{S_{p,n}} + \lambda_B B + \lambda_F \sum_i F_i - \pi^* K_{X_{p,n}}$ is exceptional and numerically trivial, hence zero by negative definiteness. Therefore $L = \pi^*(-K_{X_{p,n}}) = \Lambda$, proving the stated formulas. The SNC pullback with coefficients less than one proves that the target is klt, in every characteristic, without a tame-quotient assertion. No retained curve has square -1 , so this is the minimal resolution. Each of its $2n+1$ connected blocks contracts to a singular point: the exceptional locus of a nontrivial resolution over a smooth point would contain a (-1) -curve. Distinct blocks have distinct images by connected fibers. The sign of $d_{p,n}$ gives all three canonical-sign cases, since $(p-2)(n-2) < 2$ holds exactly in the positive cases stated. $\square$

10.1. Even sets and inseparable bisections in characteristic two. The code that occurs below is the classical doubled-even code [5, Section 2]

$$DE(n) = \{(x_1, x_1, \dots, x_n, x_n) : (x_1, \dots, x_n) \in \mathbb{F}_2^n, \sum_i x_i = 0\}.$$

It has dimension $n-1$ and minimum nonzero weight four for $n \geq 2$. The code of the exceptional curves F_i, U_i below is $DE(n)$. The same surface has an inseparable bisection.

Theorem 10.2 (Rulings and divisibility on the Keel–McKernan family). *For every algebraically closed field of characteristic two and every $n \geq 3$, the preceding construction gives a rank-one klt del Pezzo surface $X_{2,n}$ with $2n+1$ distinct singular points and*

$$K_{X_{2,n}}^2 = \frac{2}{n-2}.$$

Its minimal exceptional forest is $[2n-4] + 2nA_1$. Write $U_i = E_{i1} - E_{i2}$ and $P_i = E_{i2}$. On its original resolution,

$$L = \frac{2a + (n-1)b - \sum_i (E_{i1} + E_{i2})}{n-2}, \quad \ell_{\text{ext}} = \frac{1}{n-2}, \quad L + \ell_{\text{ext}} K_S = \frac{n-3}{n-2} b.$$

The rational ruling of class b has exactly the n reducible fibers

$$F_i + U_i + 2P_i,$$

whose exterior fiber components P_i are shortest. Its sole exceptional horizontal component is the purely inseparable bisection B . For $n \geq 4$, the complete code of two-divisible subsets of the $2n$ isolated exceptional nodes is

$$\mathcal{C}_D = \left\{ \sum_{i \in J} (F_i + U_i) : J \subseteq \{1, \dots, n\}, |J| \text{ even} \right\}, \quad \dim_{\mathbb{F}_2} \mathcal{C}_D = n - 1.$$

Here the displayed effective divisor has half-class $\frac{|J|}{2}b - \sum_{i \in J} P_i$. Its nonzero minimum support size is four. For $n = 3$, all seven exceptional curves are isolated nodes and the code has dimension three, with all seven nonzero words of weight four. In particular the seven-point bound is false in characteristic two, and the isolated-node vanishing theorem does not extend to that characteristic.

Proof. Set $p = 2$ in Proposition 10.1; this proves the construction, contraction, klt condition, Picard number, ample anticanonical divisor, singular-point count, and square. The class $(n - 2)L$ is integral and has positive integral degree on every exterior prime. Each P_i has degree $1/(n - 2)$, proving the minimum without the nef-threshold lemma or the upper-bound theorem. Substituting $K_S = -2a - 2b + \sum_i (E_{i1} + E_{i2})$ gives the stated nef adjoint. The fiber formula follows from the two blowups; all other fibers of the original ruling are unchanged and irreducible. The graph projection to the y -line is $x \mapsto x^2$, and its strict transform B is isomorphic to the original graph. Thus this is an inseparable bisection meeting P_i once in a multiplicity-two fiber. The other retained curves are vertical, so every horizontal exceptional curve has degree at most two.

For $n \geq 4$, $B^2 = 4 - 2n < -2$, leaving exactly F_i, U_i as isolated nodes. In the integral total-transform basis,

$$F_i = b - E_{i1} - E_{i2}, \quad U_i = E_{i1} - E_{i2}, \quad F_i + U_i = b - 2P_i.$$

A subset $\sum_i (\alpha_i F_i + \beta_i U_i)$, with coefficients zero or one, is twice a Picard class exactly when its coefficients in this free basis are even. This is equivalent to $\alpha_i = \beta_i$ for every i and $\sum_i \alpha_i = 0$ in $\mathbb{F}_2$. This identifies the code with $DE(n)$ in the coordinate order $(F_1, U_1, \dots, F_n, U_n)$ and proves the half-class formula; its dimension and minimum weight are the standard ones stated above. For $n = 3$ include the coefficient γ of $B = 2a + b - \sum_i (E_{i1} + E_{i2})$. The parity equations are $\beta_i = \alpha_i + \gamma$ and $\sum_i \alpha_i = \gamma$ in $\mathbb{F}_2$. They give dimension three. If $\gamma = 0$, a nonzero word uses two pairs, hence four nodes; if $\gamma = 1$, it uses B and exactly one node from each of the three pairs, again four nodes. □

Corollary 10.3 (Failure of the separability arguments in characteristic two). *Both uses of odd characteristic in the upper-bound proof fail on the preceding characteristic-two surfaces. First, the bisection has degree-two fibers at every point and zero differential, rather than two ramification points governed by a separable Riemann–Hurwitz formula. Second, for distinct indices i, j there is the explicit even set*

$$F_i + U_i + F_j + U_j \sim 2(b - P_i - P_j).$$

The degree-two algebra defined by its square-root equation is purely inseparable, and its restriction to an original rational tree disjoint from that set is a nonreduced double tree, not two disjoint copies. Thus neither the bisection ramification bound nor the even-node obstruction extends to characteristic two.

Proof. The bisection assertion follows from $y = x^2$ and the fiber formula; by Lemma 7.2, its fiber multiplicities cannot be summed as separable ramification. The even set was proved by the Picard-basis identity. At its prime components the defining section has odd valuation, so it is not a square in $k(S)$. Adjoining its square root in characteristic two therefore gives a field

extension that is purely inseparable of degree two. On a disjoint original rational tree the half-class has multidegree zero and is trivial by Lemma 2.2; the section restricts to a nonzero constant $c \in k$. The restricted algebra is

$$\mathcal{O}_T[z]/(z^2 - c) \simeq \mathcal{O}_T[\varepsilon]/(\varepsilon^2),$$

not $\mathcal{O}_T \oplus \mathcal{O}_T$. For a function-field description, take i, j to be the selected points $x = 0, 1$. Away from the exceptional locus, the square-root equation is $z^2 = y(y - 1) = y^2 + y$. Putting $w = z + y$ gives $w^2 = y$, so the extension is purely inseparable. Thus the doubled negative-definite subspace used in the proof of Theorem 5.1 is absent. $\square$

Proof of Theorem 1.2. In Proposition 10.1, take $(p, n) = (3, 3)$. The contracted exceptional divisor has four isolated (-3) -curves and three disjoint chains of two (-2) -curves. Thus it gives seven distinct singular points, and the formula there gives $K_X^2 = 1/3$. For $p = 2$ and any $n \geq 3$, the same proposition gives the family $X_{2,n}$ and all the invariants stated in Theorem 1.2. Together with Theorem 1.1, the first example proves sharpness in characteristic three. $\square$

APPENDIX A. NEGATIVE CURVES ON THE RESOLUTION OF THE SEVEN-POINT SURFACE

On the minimal resolution of the seven-point surface, the one-component replacements at curves of least positive L -degree reduce the singular-point count from seven to six.

Set $p = 3$ and choose the three graph parameters $I = \{0, 1, \infty\}$, so the surface and all the following curves are defined over $\mathbb{F}_3$. Write $S = S_3$, $D = D_3$, and

$$L = a + \frac{2}{3}b - \frac{1}{3} \sum E_{ij} = \frac{1}{3}(B + b), \quad L^2 = \frac{1}{3}.$$

When $\{i, j, k\} = I$, subscripts denote these three graph parameters. Every displayed equation denotes its bihomogeneous closure and then its strict transform on S .

Proposition A.1 (The thirteen nonexceptional negative curves on the minimal resolution). *The complete list of exterior negative prime curves on S is*

$$\{P_i, T_i, Q_i, W_i : i \in I\} \cup \{\Theta\}.$$

They are smooth rational (-1) -curves, with classes

$$\begin{aligned} T_i &= a - E_{i1}, & \Theta &= a + b - \sum_i E_{i1}, \\ Q_i &= 2a + b - E_{i1} - \sum_{j \neq i} (E_{j1} + E_{j2}), \\ W_i &= 3a + 2b - (E_{i1} + E_{i2} + E_{i3}) - \sum_{j \neq i} (2E_{j1} + E_{j2}). \end{aligned}$$

Here T_i is the strict a -fiber through the selected point, and the remaining equations are

$$\begin{array}{l|l} \Theta & x - y = 0 \\ Q_0 & x^2 + x + y = 0 \\ Q_1 & x^2 - y = 0 \\ Q_\infty & x^2 + xy + y = 0 \end{array} \quad \begin{array}{l|l} W_0 & x^3 + y^2 + y = 0 \\ W_1 & x^3 - y^2 = 0 \\ W_\infty & x^3y + x^3 + y^2 = 0. \end{array}$$

Their L -degrees are respectively

$$L \cdot P_i = L \cdot T_i = \frac{1}{3}, \quad L \cdot \Theta = L \cdot Q_i = \frac{2}{3}, \quad L \cdot W_i = 1.$$

In particular the six shortest exterior curves are exactly P_i, T_i , and the least exterior degree is $1/3$, established without using Lemma 3.3 or the seven-point bound. The surface has exactly twenty-three negative prime curves including D .

Proof. The charts in the preceding proposition construct P_i, T_i . The diagonal and the three Q_i are graphs of morphisms $\mathbb{P}^1 \rightarrow \mathbb{P}^1$. For Q_1 , the graph $y = x^2$ has cluster multiplicities $(1, 1, 0)$ at $0, \infty$ and $(1, 0, 0)$ at 1 . The diagonal has $(1, 0, 0)$ at all three points. The curve W_1 has normalization $x = z^2, y = z^3$ and only its two cusps at $0, \infty$; each cusp is resolved by its first blowup. Its cluster multiplicities are $(2, 1, 0)$ at those points and $(1, 1, 1)$ at 1 . For example at

0, after writing $v = uv_1$ its strict equation is $u = v_1^2$. Blowing up its common point with the strict graph, in the other chart $u = v_1u_2$, gives $u_2 = v_1$; this is smooth through the node of U_0 and V_0 . The third graph blowup is at a different point. At 1, the equations after the three graph substitutions are $u^2 + v_1 - uv_1^2$, $u + v_2 - u^2v_2^2$, and $1 + v_3 - u^3v_3^2$. The last meets P_1 at $v_3 = -1$, distinct from the three original contacts. Reciprocal coordinates prove the same cusp calculation at infinity. The simultaneous transformations $(x, y) \mapsto (g(x), g(y))$ for $g(z) = 1 - z$ and $g(z) = z/(z - 1)$ preserve the Frobenius graph and permute $0, 1, \infty$. They lift through the entire blowup configuration. They carry the Q_1, W_1 calculations to the other two rows. Thus every listed strict transform is smooth and rational. The class formulas and intersection form give self-intersection -1 and the displayed L -degrees.

We prove completeness independently of those constructions. Every exterior prime Z has $K_S \cdot Z < 0$; if $Z^2 < 0$, adjunction makes it a smooth rational (-1) -curve. Since $3L$ is integral and all four higher-weight coefficients are $1/3$, write $L \cdot Z = r/3$ with $r = 1, 2, 3$. Let z_0, z_i be its contacts with B, F_i , and u_i, v_i its contacts with U_i, V_i . The discrepancy-sum and projection identities give

$$z_0 + \sum_i z_i = 3 - r, \quad z_0^2 + \sum_i z_i^2 + 2 \sum_i (u_i^2 + u_i v_i + v_i^2) = 3 + r^2.$$

The negative matrix of D is the sum of four $[3]$ blocks and three $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ blocks. Solving its rank-one closure gives the coefficients of $[Z]$ in the integral basis a, b, E_{ij} :

$$\begin{aligned} [a]Z &= r - z_0, & [b]Z &= r - 1, \\ [E_{i1}]Z &= (-r + z_0 + z_i - 2u_i - v_i)/3, \\ [E_{i2}]Z &= (-r + z_0 + z_i + u_i - v_i)/3, \\ [E_{i3}]Z &= (-r + z_0 + z_i + u_i + 2v_i)/3. \end{aligned}$$

In particular $z_0 + z_i - r + u_i - v_i \equiv 0 \pmod{3}$ for each i . For $r = 1$, a single higher-weight contact two violates integrality at an untouched branch. Otherwise two higher-weight contacts are one and exactly one weight-two contact is one. At the two untouched branches integrality forces $z_j = 1 - z_0$; at the touched branch it forces $z_i = z_0$. For $z_0 = 0$ the weight-two contact is U_i , giving T_i ; for $z_0 = 1$ it is V_i , giving P_i . For $r = 2$, exactly one higher-weight contact is one and $\sum(u_i^2 + u_i v_i + v_i^2) = 3$. The congruences force all $u_i = 1, v_i = 0$ when $z_0 = 1$, giving Θ ; when $z_i = 1$ they force $u_i = 1, v_i = 0$ and $u_j = 0, v_j = 1$ for $j \neq i$, giving Q_i . For $r = 3$ there are no higher-weight contacts. The quadratic equation gives $u_i, v_i \leq 2$, and integrality forces $u_i = v_i$. There must therefore be two pairs $(1, 1)$ and one pair $(0, 0)$, giving exactly W_i at the untouched branch. All thirteen resulting integral classes were realized above. Two distinct prime curves cannot share a negative numerical class, since their nonnegative intersection would equal that negative square. This proves completeness. $\square$

Proposition A.2 (Adjoint identities and singular-point counts on the equality example). *For $\{i, j, k\} = I$ the following are linear equivalences between actual integral divisors on the unchanged S :*

$$\begin{aligned} K_S + V_i + B + F_i + 2P_i &\sim T_i, \\ K_S + U_i + F_j + F_k + 2T_i &\sim P_i, \\ K_S + \sum_i U_i + 2\Theta &\sim \sum_i P_i, \\ K_S + U_i + V_j + V_k + 2Q_i &\sim P_i + T_j + T_k, \\ K_S + U_j + V_j + W_i &\sim P_k + T_k \quad (j \neq i). \end{aligned}$$

The right sides are the unique effective representatives of their classes. The six shortest curves thus form three degree-preserving adjoint pairs $P_i \leftrightarrow T_i$; the other seven exterior (-1) -curves have displayed adjoint reductions to shortest curves. At P_i , the unique excess-contact vertex is V_i , of valency one in D ; at T_i it is U_i , also of valency one in D . The corresponding one-component replacement applies, but changes the singular-point count by $1 - 2 = -1$. Its target

has six singular points, minimal-resolution exceptional forest $3A_2 + A_1 + 2[3]$, and anticanonical square $2/3$. Thus these six one-component replacements do not provide a singular-point preserving Picard descent.

Proof. The complete shortest contact rows are

curve	original contacts, all once	excess-contact vertex
P_i	B, F_i, V_i	V_i
T_i	F_j, F_k, U_i	U_i

The blowup charts give these intersections. Substitution of the proved curve classes and $K_S = -2a - 2b + \sum E_{ij}$ gives all five adjoint identities. Each right side consists of pairwise disjoint (-1) -curves with coefficient one. Intersection with each of them forces it to occur in any effective representative, and the remaining effective divisor would be linearly trivial, hence zero. This proves the uniqueness assertions. For the last identity the curve W_i passes through the exceptional node $U_j \cap V_j$ as a third smooth branch; the adjacent-contact adjoint allows exactly this coincidence, so no distinct-point hypothesis is used.

Delete V_i at P_i and blow down P_i . The original B and F_i become transverse (-2) -curves forming one A_2 block. The curve U_i remains a separate A_1 block, the two other A_2 blocks are unchanged, and the other two F -curves remain isolated of square -3 . All contacts on P_i were distinct, so these are the images. The original retained support $P_i + (D - V_i)$ is an SNC forest and all remaining contacts are bounded. Thus the one-component replacement applies with $h = \eta = 2/3$, $c = 1$; its nonminimal anticanonical pullback is

$$L_i^\dagger = -K_S + P_i - \frac{1}{3}(F_j + F_k).$$

Its square is $2/3$, and the retained minimal-resolution configuration is the forest with six connected components just listed. The component change is $s_{V_i} - 2 = -1$. At T_i delete U_i and blow down T_i instead. The curves F_j, F_k become the new A_2 , V_i is the isolated A_1 , and B, F_i are the two remaining $[3]$ blocks. The nonminimal pullback is

$$\tilde{L}_i^\dagger = -K_S + T_i - \frac{1}{3}(B + F_i), \quad (\tilde{L}_i^\dagger)^2 = \frac{2}{3}.$$

This proves the same conclusions. In both cases the contraction theorem inside Theorem 4.5 verifies projectivity, klt singularities, Picard number one, and ample anticanonical divisor of the target. □

Corollary A.3 (Intersection lattices at equality). *The seven-point bound of Theorem 1.1 is optimal. The sharp surface has $K_S^2 = -1$, ten original exceptional components, three original edges, $\det A_D = 3^7$, $\ell = L^2 = 1/3$, and at every shortest center*

$$p^\top A_D^{-1} p = \frac{4}{3}, \quad |\det \langle D, P \rangle| = 3^6, \quad [\text{Pic}(S) : \langle D, P \rangle] = 27.$$

It has no isolated exceptional nodes. Thus its full Picard index is integral, and there is no isolated-node summand to which Lemma 5.2 could apply. For $\beta = 3$, the single-adjoint table allows at most two edges, as required by the assumption of at least eight singular points; the example has three.

Proof. The contraction and seven singular points are proved in Proposition 10.1; together with the independently proved upper bound this gives optimality. The original matrix has four $[3]$ blocks and three A_2 blocks, giving determinant 3^7 and no isolated A_1 component. The projection and full-index formulas give the displayed values. Finally $10 - 3 = 7$ is the component count of the exceptional forest. □

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